FLAT TOP LOAD CHARTS

PART 1 RADIUS AND CAPACITY

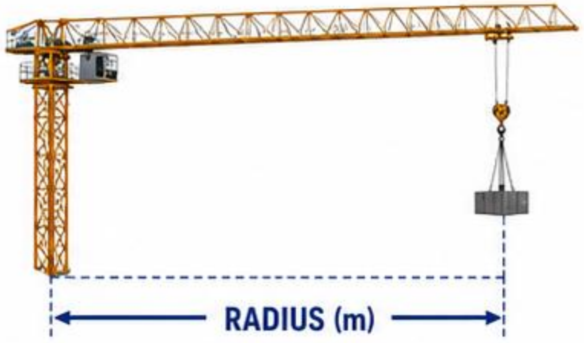
LEARN TO READ THE CHART.
KNOW THE LIMITS. LIFT SAFELY.

COMEDIL

CTT 561-20

WHAT IS RADIUS?

Radius is the horizontal distance from the center of rotation (center of the mast) to the center of the load.



HOW TO FIND CAPACITY

1. Select the correct jib length that is installed.
2. Find your working radius (in meters) along the top row.
3. Read down to the 2-Part Line (or 4-Part Line) for capacity in pounds (lbs).

CTT 561-20
85 m JIB – 2 PART LINE

RADIUS (m)	20	25	30	35	40	45	50	55	60	65	70	75	80	85
LBS	22,920	19,070	16,210	14,330	12,700	11,400	10,300	9,380	8,600	7,980	7,440	6,950	6,520	6,120

At 40 m Radius = 12,700 lbs (2-Part Line)

RADIUS IS HORIZONTAL

Always measure from the center of the crane (slew ring) to the center of the load.

CAPACITY DECREASES

As radius increases, the capacity the crane can safely lift decreases.

USE CHARTED VALUES

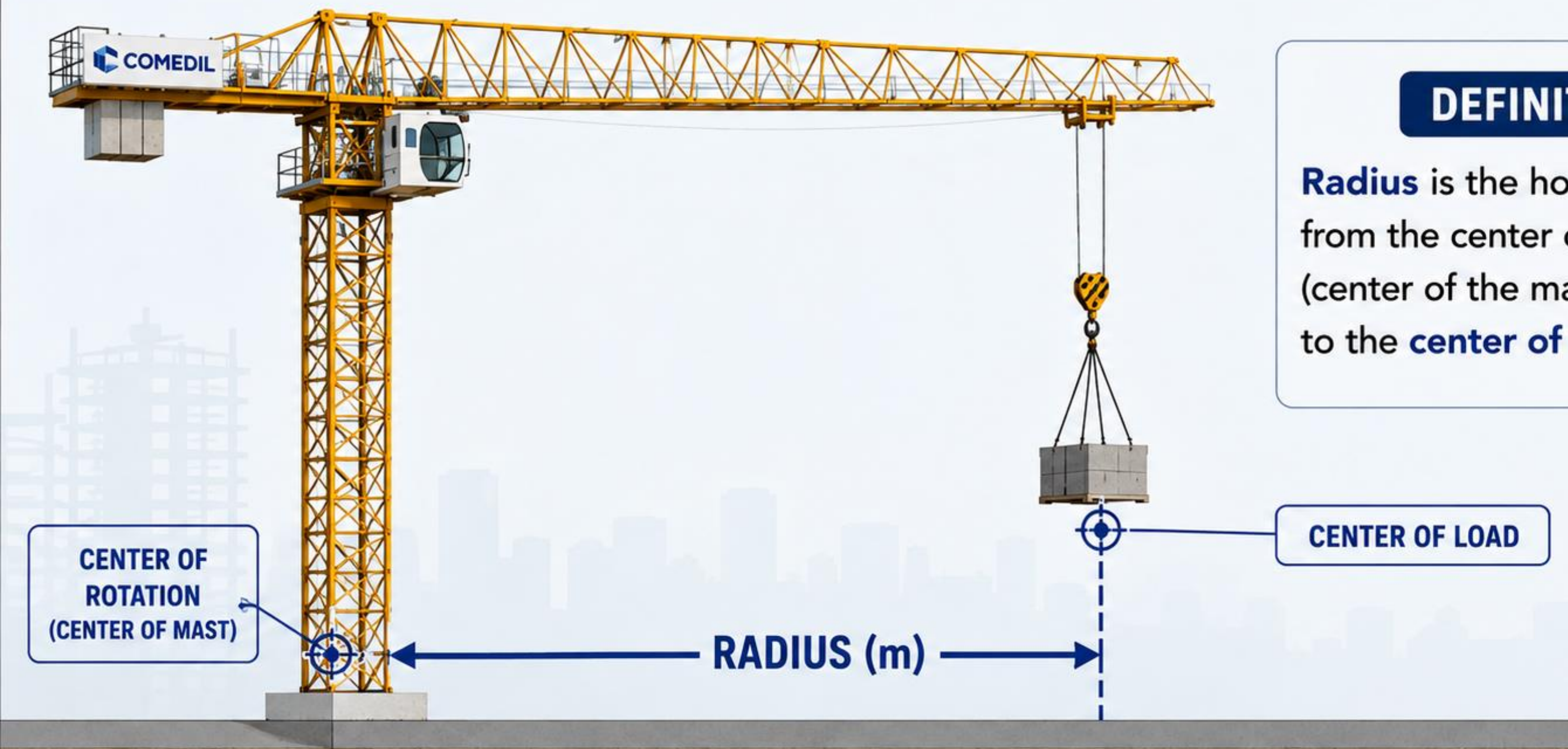
Never guess or interpolate. Always use the capacity shown in the chart.

RESPECT THE LIMITS

Exceeding charted capacity can cause serious injury, fatality and equipment failure.

SAFE CRANES. SAFE LIFTS. SAFE PEOPLE.

1. WHAT IS RADIUS?



DEFINITION

Radius is the horizontal distance from the center of rotation (center of the mast) to the **center of the load.**

Always measure radius in **METERS** (m).

2. HOW TO FIND CAPACITY

1 SELECT JIB LENGTH
Use the load chart that matches your installed jib length.

2 FIND WORKING RADIUS
Locate your radius (in meters) along the top row.

3 READ CAPACITY
Read down to the correct part line (2-Part or 4-Part) for the capacity in pounds (lbs).

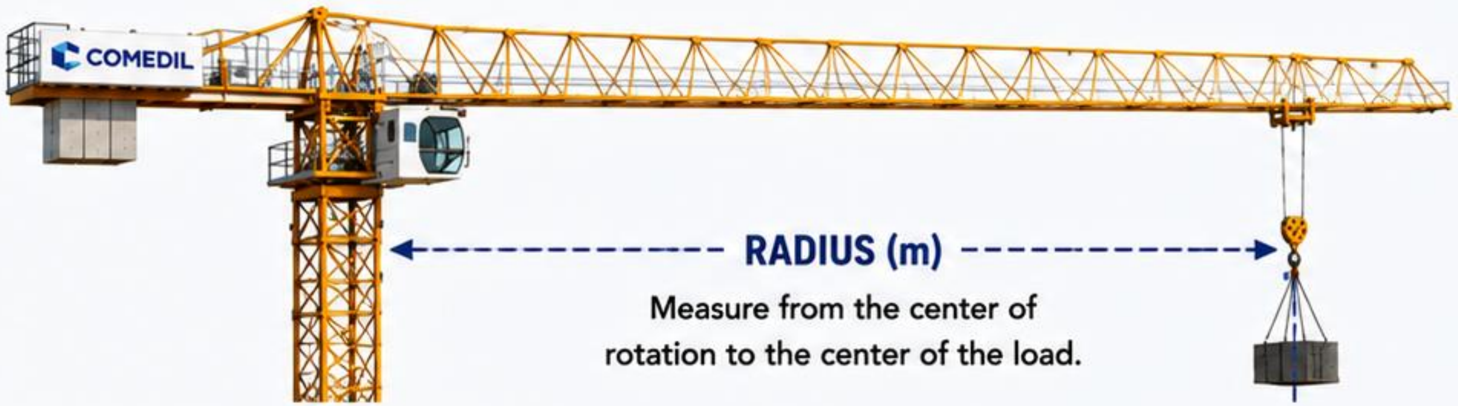
LBS

ALWAYS USE THE VALUE IN THE CHART.
Never guess or interpolate.

EXAMPLE: CTT 561-20 • 85 m JIB – 2 PART LINE

RADIUS (m)	20	25	30	35	40	45	50	55	60	65	70	75	80	85
LBS (2-Part Line)	22,920	19,070	16,210	14,330	12,700	11,400	10,300	9,380	8,600	7,980	7,440	6,950	6,950	6,120

At 40 m Radius on 2-Part Line = **12,700 lbs**



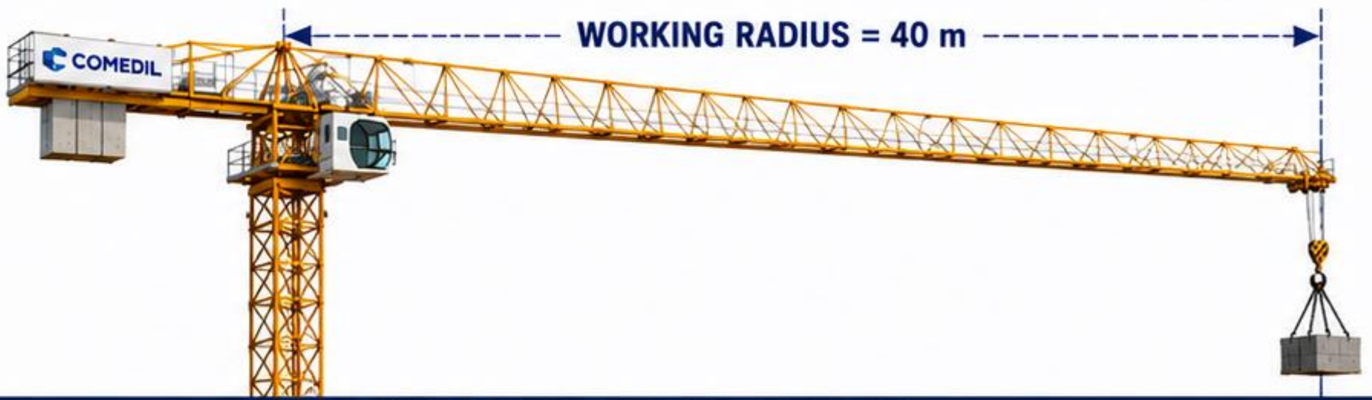
READ THE NOTES! Notes at the bottom of the load chart can affect capacity.
They tell you what is INCLUDED or EXCLUDED.

- Line allowance
- Reeving (2-Part / 4-Part)
- Hook block / ball deductions
- Units (lbs or kg) and configuration

3. GROSS LOAD CALCULATION

Gross Load is the TOTAL weight on the hook.

EXAMPLE: CTT 561-20 • 85 m JIB – 2 PART LINE



1 FIND CHART CAPACITY

Locate your radius and read the capacity from the appropriate part line.

2 CALCULATE GROSS LOAD

Add the weight of the load and all rigging/attachments on the hook.

3 COMPARE

Gross Load must be LESS THAN or EQUAL TO the chart capacity.

SAFE IF:

Gross Load (Total on Hook)
≤ **Chart Capacity**

FROM THE LOAD CHART (2-PART LINE)

RADIUS (m)	20	25	30	35	40	45	50
CAPACITY (lbs) (2-Part Line)	22,920	19,070	16,210	14,330	12,700	11,400	10,300

EXAMPLE CALCULATION

Load Weight (example)	=	18,000 lbs
+ Rigging & Attachments (example)	=	2,500 lbs
GROSS LOAD (Total on Hook)	=	**20,500 lbs**

Chart Capacity at 40 m (2-Part Line)
= **12,700 lbs**

20,500 lbs > 12,700 lbs
NOT SAFE

⚠ **NEVER EXCEED CHART CAPACITY.** Reduce the load, decrease radius, or use more parts of line.

4. BETWEEN RADII – FINDING CAPACITY

When your radius is between two values on the chart.

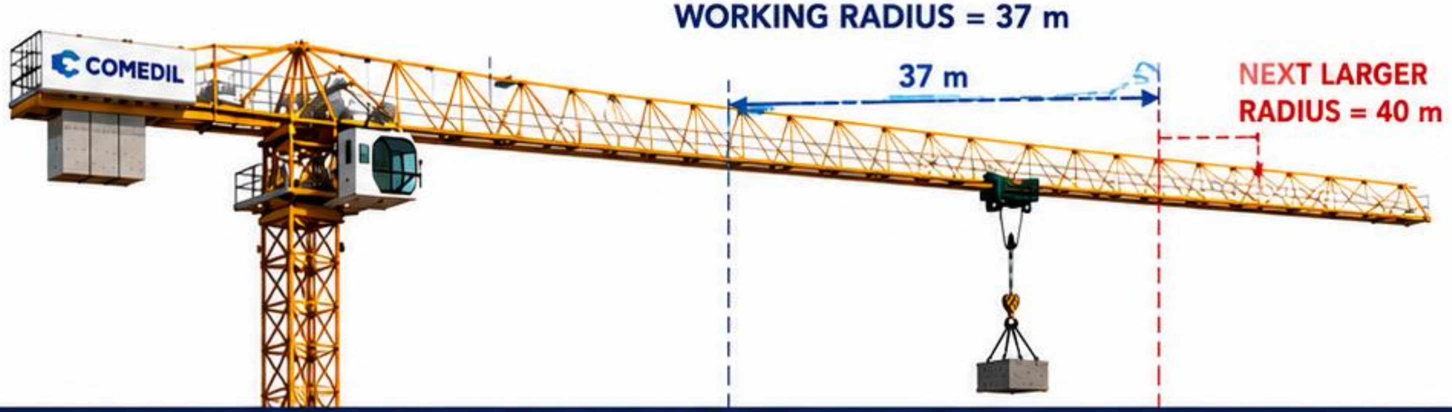
THE RULE:

When your working radius falls between two radii on the chart, use the **NEXT LARGER** radius.

WHY?

Capacity decreases as radius increases. The next larger radius gives you a conservative (safe) value you know the crane can lift.

EXAMPLE: CTT 561-20 • 85 m JIB • 2 PART LINE



FROM THE LOAD CHART (2-PART LINE)

RADIUS (m)	30	35	37	40	45	50
CAPACITY (lbs)	16,210	14,330	—	12,700	11,400	10,300

QUESTION

The working radius is 37 m (2-Part Line).

What is the chart capacity?

SOLUTION

1. 37 m falls between 35 m and 40 m on the chart.
2. Use the **NEXT LARGER** radius – 40 m.

CHART CAPACITY = 12,700 lbs (at 40 m)

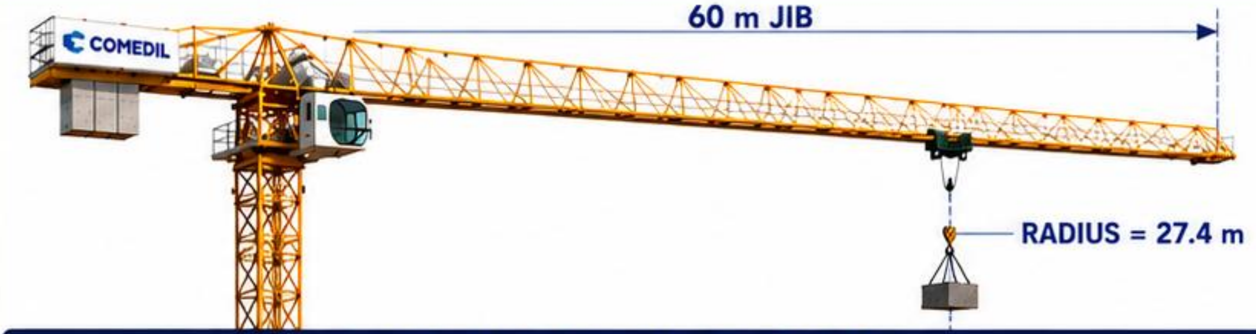
⚠ **NEVER EXCEED CHART CAPACITY.** Reduce the load, decrease radius, or use more parts of line.

5. JIB LENGTH CONFIGURATIONS

Always use the correct load chart for the actual jib length in use.

A tower crane's capacity depends on the jib length. Different jib lengths = different load charts.

EXAMPLE 1: 60 m JIB CONFIGURATION

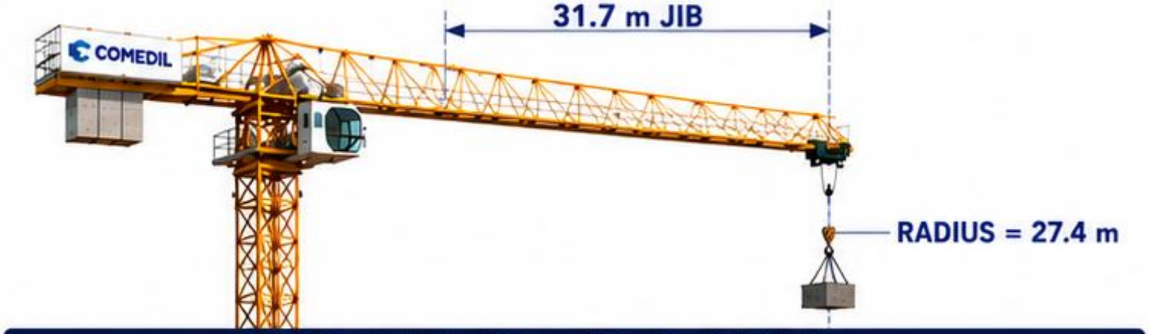


LOAD CHART – 60 m JIB – 2 PART LINE

RADIUS (m)	15	20	25	27.4	30	35	40	45	50	55	60
CAPACITY (kg)	16,700	12,900	10,400	9,490	8,200	5,950	4,460	3,560	2,900	2,350	1,950

At 27.4 m radius (60 m jib): CAPACITY = **9,490 kg**

EXAMPLE 2: 31.7 m JIB CONFIGURATION



LOAD CHART – 31.7 m JIB – 2 PART LINE

RADIUS (m)	15	20	25	27.4	30
CAPACITY (kg)	9,620	7,400	6,150	5,950	5,220

At 27.4 m radius (31.7 m jib): CAPACITY = **5,950 kg**

QUESTION

The crane is set up at a working radius of 27.4 m (2-part line).

Which configuration provides the greater lifting capacity?

A) 60 m jib
B) 31.7 m jib
C) Same capacity

SOLUTION

1. Use the load chart for the actual jib length in use.
2. Find 27.4 m on each chart (or the next larger radius if not listed).
3. Read and compare the capacities.

60 m jib capacity at 27.4 m	=	**9,490 kg**
31.7 m jib capacity at 27.4 m	=	**5,950 kg**

ANSWER: A) 60 m jib has the greater capacity.

KEY REMINDERS

- **Always use the load chart that matches the actual jib length in use.**
- **Capacity decreases as radius increases.** The further out, the less you can lift.
- **Never exceed the chart capacity. Plan the lift and stay within limits.**

NEVER EXCEED CHART CAPACITY. Reduce the load, decrease radius, or use more parts of line.

6. NET LOAD vs. GROSS LOAD

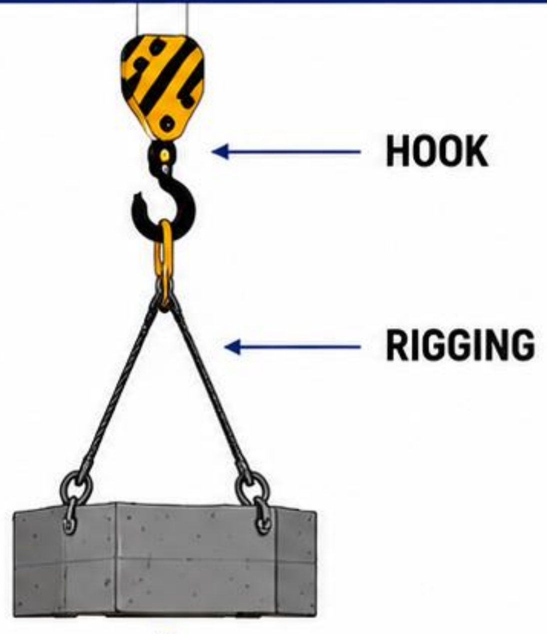
The load chart uses **GROSS LOAD** (total weight on the hook).

NET LOAD

The weight of the load only.

Example:

17,250 lb



NET LOAD
(Load Only)

GROSS LOAD

The load **PLUS** all rigging and attachments suspended from the hook.

Example:

17,250 lb
\+ 170 lb rigging

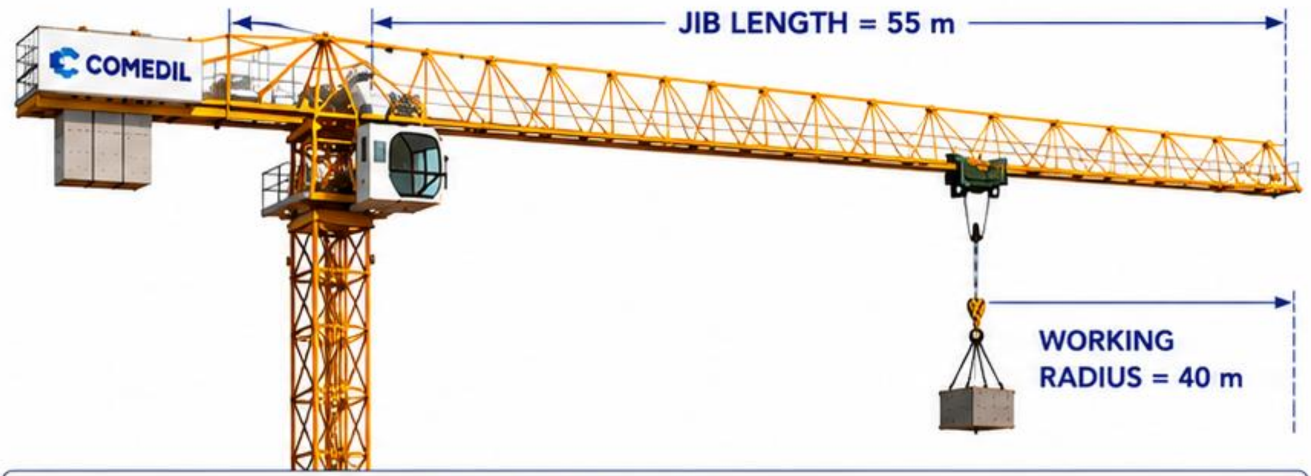
= 17,420 lb GROSS

GROSS LOAD
(Total on Hook)

THE LOAD CHART QUESTION USES GROSS LOAD — NOT JUST THE LOAD WEIGHT.

7. GROSS CAPACITY vs. NET CAPACITY

Understand the difference between what the chart shows (Gross) and what is available for the load (Net).



GROSS CAPACITY (Chart Capacity)

The total capacity the crane can lift at a given radius and jib length.

This is the number shown in the load chart.

EXAMPLE

From the load chart at 55 m jib and 40 m radius:

12,700 lb
GROSS CAPACITY

NET CAPACITY (Available for the Load)

The capacity available for your load after subtracting the weight of all rigging and attachments.

EXAMPLE

Gross Capacity (from chart)	12,700 lb
– Rigging (2-Part Line)	170 lb
= Net Capacity (available)	12,530 lb

THE RELATIONSHIP

GROSS CAPACITY (chart) – RIGGING + ATTACHMENTS = NET CAPACITY (available)

GROSS CAPACITY

- Comes from the load chart.
- Based on radius, jib length, and parts of line.
- Includes the weight of the hook, block, slings, rigging, and load.

NET CAPACITY

- What remains for the actual load.
- Use this number to make sure your load (NET) is within limits.
- Never let your load exceed the Net Capacity.

TYPICAL RIGGING WEIGHT EXAMPLES

2-PART LINE	4-PART LINE	TAG LINE	CHAINS / OTHERS
~ 100 – 200 lb	~ 200 – 300 lb	~ 10 – 20 lb	Varies

ALWAYS USE NET CAPACITY TO CHECK YOUR LOAD. | **NEVER EXCEED THE GROSS CAPACITY.**

The load chart gives you GROSS CAPACITY. | After deductions, what you have left is NET CAPACITY. | Stay within the Net Capacity and lift safely.

PRACTICE QUESTION 1

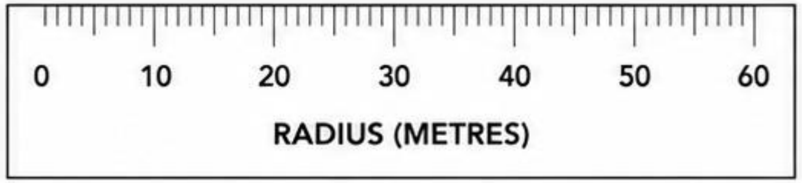
READING CAPACITY – 2-PART LINE

The crane is set up with a **55 m** jib.

The working radius is **40 m**.

Using **2-part line**.

What is the chart capacity?



LIEBHERR 200 HC

Max Load: 20,000 kg
Tip Load: 2,700 kg

Tower Height: 51.4 m
Max Jib: 120 ft / 36.6 m

2-Part Line Lift Rope: 22 mm
4-Part Line Lift Rope: 22 mm

LOAD CHART – 2-PART LINE (kg)

RADIUS (m)	15	20	25	30	35	40	45	50	55	60
JIB CONFIGURATION										
60 m / 197 ft	20,000	20,000	18,930	15,560	12,600	10,310	8,450	6,950	5,780	4,860
55 m / 180 ft	20,000	20,000	18,930	15,560	12,600	10,200	8,370	6,890	5,720	—
50 m / 164 ft	20,000	20,000	18,930	15,560	12,500	10,090	8,270	6,790	—	—
45 m / 148 ft	20,000	20,000	18,930	15,560	12,400	10,010	8,190	—	—	—
40 m / 131 ft	20,000	20,000	18,940	15,560	12,300	9,930	—	—	—	—
35 m / 115 ft	20,000	20,000	18,950	15,600	12,200	—	—	—	—	—
30 m / 98 ft	20,000	20,000	18,970	15,650	—	—	—	—	—	—

Note: Capacities in kilograms. — = beyond chart radius.
Always use the chart that matches the actual jib length in use.

SOLUTION – PRACTICE QUESTION 1

1 Identify the correct chart.

Jib length = **55 m**

Use the **55 m / 180 ft** row.

2 Move across to the working radius.

Radius = **40 m**

Move down to the **40 m** column.

3 Read the chart capacity.

Where the **55 m** row and **40 m** column intersect is **10,200 kg.**

CHART CAPACITY
10,200 kg

REMEMBER

- Capacities are in kilograms.
- Use the chart that matches the actual jib length.
- 2-part line is shown in this chart.

LIEBHERR 200 HC

Max Load: 20,000 kg
Tip Load: 2,700 kg

Tower Height: 51.4 m
Max Jib: 120 ft / 36.6 m

2-Part Line Lift Rope: 22 mm
4-Part Line Lift Rope: 22 mm

LOAD CHART – 2-PART LINE (kg)

RADIUS (m)	15	20	25	30	35	40	45	50	55	60
JIB CONFIGURATION										
60 m / 197 ft	20,000	20,000	18,930	15,560	12,600	10,310	8,450	6,950	5,780	4,860
55 m / 180 ft	~~20,000~~	~~20,000~~	~~18,930~~	~~15,560~~	~~12,600~~	**10,200**	8,370	6,890	5,720	—
50 m / 164 ft	20,000	20,000	18,930	15,560	12,500	10,090	8,270	6,790	—	—
45 m / 148 ft	20,000	20,000	18,930	15,560	12,400	10,010	8,190	—	—	—
40 m / 131 ft	20,000	20,000	18,940	15,560	12,300	9,930	—	—	—	—
35 m / 115 ft	20,000	20,000	18,950	15,600	12,200	—	—	—	—	—
30 m / 98 ft	20,000	20,000	18,970	15,650	—	—	—	—	—	—

Note: Capacities in kilograms. — = beyond chart radius.
Always use the chart that matches the actual jib length in use.

TAKEAWAY: At **55 m** jib, **40 m** radius, using **2-part line**, the chart capacity is **10,200 kg**.

PRACTICE QUESTION 2

A load is being lifted with the following rigging and attachments.

LOAD INFORMATION	
Load (weight of load)	17,250 lb
Weight of rigging and attachments	170 lb

What is the gross load?

(Do not round your answer.)

LIEBHERR 200 HC

Max Load: 20,000 kg
Tip Load: 2,700 kg
Tower Height: 51.4 m
Max Jib: 120 ft / 36.6 m
2-Part Line Lift Rope: 22 mm
4-Part Line Lift Rope: 22 mm

LOAD CHART – 2-PART LINE (kg)

RADIUS (m)	15	20	25	30	35	40	45	50	55	60
JIB CONFIGURATION										
60 m / 197 ft	20,000	20,000	18,930	15,560	12,600	10,310	8,450	6,950	5,780	4,860
55 m / 180 ft	20,000	20,000	18,930	15,560	12,600	10,200	8,370	6,890	5,720	—
50 m / 164 ft	20,000	20,000	18,930	15,560	12,500	10,090	8,270	6,790	—	—
45 m / 148 ft	20,000	20,000	18,930	15,560	12,400	10,010	8,190	—	—	—
40 m / 131 ft	20,000	20,000	18,940	15,560	12,300	9,930	—	—	—	—
35 m / 115 ft	20,000	20,000	18,950	15,600	12,200	—	—	—	—	—
30 m / 98 ft	20,000	20,000	18,970	15,650	—	—	—	—	—	—

Note: Capacities in kilograms. — = beyond chart radius.
Always use the chart that matches the actual jib length in use.

SOLUTION – PRACTICE QUESTION 2

We need to find the **GROSS LOAD.**

Gross load is the total weight the crane must lift, which includes the load plus all rigging and attachments.

CALCULATION

Load (weight of load)	17,250 lb
Weight of rigging and attachments	+ 170 lb
GROSS LOAD	**17,420 lb**

TAKEAWAY:
Gross load = Load + Weight of rigging and attachments.

LIEBHERR 200 HC

Max Load: 20,000 kg
Tip Load: 2,700 kg
Tower Height: 51.4 m
Max Jib: 120 ft / 36.6 m
2-Part Line Lift Rope: 22 mm
4-Part Line Lift Rope: 22 mm

LOAD CHART – 2-PART LINE (kg)

RADIUS (m)	15	20	25	30	35	40	45	50	55	60
JIB CONFIGURATION										
60 m / 197 ft	20,000	20,000	18,930	15,560	12,600	10,310	8,450	6,950	5,780	4,860
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30 m / 98 ft	20,000	20,000	18,970	15,650	—	—	—	—	—	—

Note: Capacities in kilograms. — = beyond chart radius.
Always use the chart that matches the actual jib length in use.

ANSWER: The gross load is **17,420 lb.**

PRACTICE QUESTION 3

COMEDIL CTT 561-20

Jib length: **85 m**

Working radius: **52 m**

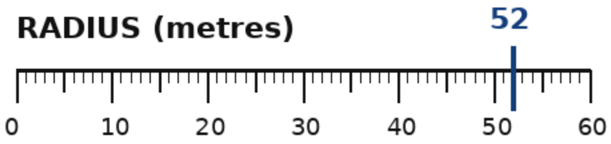
Parts of line: **2-part**

Rigging weight: **180 kg**

What is the

GROSS CAPACITY

of the crane at this radius?



COMEDIL CTT 561-20 — 2-PART LINE

Gross capacities shown in kilograms (kg)

JIB CONFIGURATION	HOOK RADIUS (m)			
Jib length	**45**	**50**	**55**	**60**
85 m	**5,650**	**4,980**	**4,430**	**3,970**
80 m	7,160	6,340	5,670	5,120
75 m	8,670	7,690	6,900	6,240
70 m	10,000	9,040	8,140	7,380
65 m	10,000	9,770	8,790	7,980
60 m	10,000	10,000	9,580	8,700
55 m	10,000	10,000	10,000	—

Use the chart as supplied. The question asks for GROSS CAPACITY.

Do not calculate net capacity unless the question asks for it.

SOLUTION – PRACTICE QUESTION 3

GIVEN

Jib length:	**85 m**
Actual radius:	**52 m**
Parts of line:	**2-part**
Rigging weight:	**180 kg**

1 **52 m is not listed.**
Move to the next larger
listed radius: 55 m.

2 **Read the 85 m jib row at the 55 m column.**

GROSS CAPACITY
4,430 kg

Rigging weight is not used.

COMEDIL CTT 561-20 — 2-PART LINE

Fulford chart values • capacities in kilograms

JIB CONFIGURATION	HOOK RADIUS (m)			
Jib length	**45**	**50**	**55**	**60**
85 m	5,650	4,980	**4,430**	3,970
80 m	7,160	6,340	5,670	5,120
75 m	8,670	7,690	6,900	6,240
70 m	10,000	9,040	8,140	7,380
65 m	10,000	9,770	8,790	7,980
60 m	10,000	10,000	9,580	8,700
55 m	10,000	10,000	10,000	—

52 m → use 55 m column → 85 m jib row → 4,430 kg

The 180 kg rigging weight is irrelevant because the question asks for GROSS CAPACITY.

PARTS OF LINE

Parts of line refers to the **number of rope sections** that support the load through the travelling block (hook block).

The more parts of line, the more rope sections share the load.

SUPPORTING ROPE SECTIONS

These rope sections support the travelling block and the load.

Top (Boom) Sheaves

Rope is reeved over the top sheaves.

Travelling Block (Hook Block)

The block moves up and down.

Hook

The load is attached to the hook.

KEY TAKEAWAY

Count the number of rope sections that support the travelling block.
That number is the parts of line.

PART 2 – PARTS OF LINE

2-PART LINE

2 PARTS OF LINE

In a 2-part line, there are **two rope sections** supporting the travelling block (hook block).

WHAT THIS MEANS:

HIGHER SPEED
Less rope is moving, so the load can be lifted faster.

LOWER MAXIMUM HOIST CAPACITY
Fewer rope sections share the load, so the maximum capacity is lower.

2
SUPPORTING ROPE SECTIONS
These two rope sections support the travelling block and the load.

TOP (BOOM) SHEAVES
The rope is reeved over the top sheaves.

TRAVELLING BLOCK (HOOK BLOCK)
Moves up and down to lift or lower the load.

HOOK
The load is attached to the hook.

KEY TAKEAWAY | 2 parts of line = faster hoist speed, lower maximum hoist capacity.

PART 2 – PARTS OF LINE

2-PART LINE vs 4-PART LINE

2-PART LINE

In a 2-part line, **two rope sections** support the travelling block (hook block).

HIGHER SPEED

Less rope is moving, so the load can be lifted faster.

LOWER MAXIMUM HOIST CAPACITY

KG

Fewer rope sections share the load, so the maximum capacity is lower.

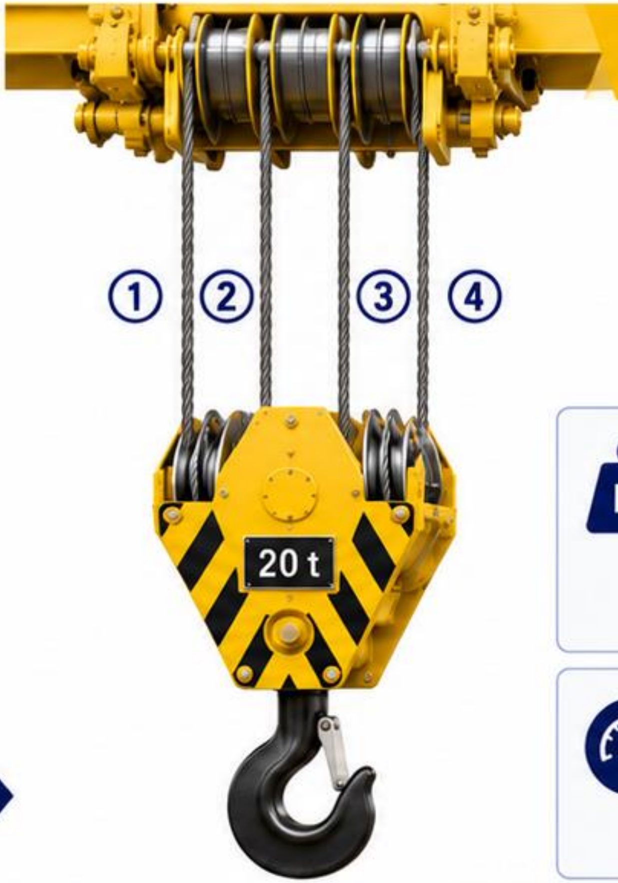
SAME CRANE
SAME JIB
SAME RADIUS

- Same crane model
- Same jib configuration
- Same radius
- Only the parts of line are different

DIFFERENT CAPACITY & SPEED

4-PART LINE

In a 4-part line, **four rope sections** support the travelling block (hook block).



HIGHER MAXIMUM HOIST CAPACITY

KG

More rope sections share the load, so the crane can lift more.

LOWER SPEED

More rope must move, so the load travels up and down slower.

EXAMPLE (COMEDIL CTT 561-20)	Hoist Gear	2-PART LINE		4-PART LINE		Difference	
		Max Gross Capacity	Hoist Speed	Max Gross Capacity	Hoist Speed		
	Gear 1	10,000 kg	46 m/min	20,000 kg	23 m/min	+10,000 kg (2X capacity)	-23 m/min (½ the speed)

KEY TAKEAWAY: More parts of line trade **speed** for **capacity**. Choose the reeving that matches the lift requirements.

Same crane. Same jib. Same radius.

The parts of line can change the available gross capacity.

COMEDIL CTT 561-20	4-PART LINE — capacity kg					
Radius (m)	**5**	**10**	**15**	**20**	**25**	**30**
55 m jib	20,000	20,000	**20,000**	20,000	20,000	20,000

COMEDIL CTT 561-20	2-PART LINE — capacity kg					
Radius (m)	**5**	**10**	**15**	**20**	**25**	**30**
55 m jib	10,000	10,000	**10,000**	10,000	10,000	10,000

AT 15 m RADIUS

55 m JIB

2-PART **10,000 kg**

4-PART **20,000 kg**

Difference: **10,000 kg**

At this short radius, 4-part **provides the higher capacity.**

READ THE CORRECT PARTS-OF-LINE SECTION.

55 m jib + 15 m radius: 2-part = 10,000 kg | 4-part = 20,000 kg

At a long radius, 4-part is not automatically the greater chart capacity.

Use the manufacturer's chart for the actual parts of line, jib length, and radius.

COMEDIL CTT 561-20			4-PART LINE — capacity kg		
Radius (m)	**65**	**70**	**75**	**80**	**85**
85 m jib	2,640	2,310	2,030	1,780	**1,600**

COMEDIL CTT 561-20			2-PART LINE — capacity kg		
Radius (m)	**65**	**70**	**75**	**80**	**85**
85 m jib	3,580	3,250	2,970	2,720	**2,500**

AT 85 m RADIUS

85 m JIB

4-PART **1,600 kg**

2-PART **2,500 kg**

Difference: **+900 kg**

At the farthest radius, the 2-part **chart shows the higher capacity.**

TRAP: MORE PARTS OF LINE DOES NOT ALWAYS MEAN MORE CAPACITY AT EVERY RADIUS.

Read the exact chart value: 85 m jib + 85 m radius → 4-part = 1,600 kg, 2-part = 2,500 kg.

FULFORD-STYLE
LOAD CHART QUESTION

Crane:	**Comedil CTT 561-20**
Jib:	**55 m**
Radius:	**15 m**
Reeving:	**4-part line**

QUESTION

What is the crane's

GROSS CAPACITY?

Both 2-part and 4-part chart sections are shown.

COMEDIL CTT 561-20 — CAPACITY EXCERPTS

Jib configurations are rows. Hook radius values are columns. Capacities are kg.

COMEDIL CTT 561-20	4-PART LINE — capacity kg					
Radius (m)	**5**	**10**	**15**	**20**	**25**	**30**
55 m jib	20,000	20,000	20,000	20,000	20,000	20,000

COMEDIL CTT 561-20	2-PART LINE — capacity kg					
Radius (m)	**5**	**10**	**15**	**20**	**25**	**30**
55 m jib	10,000	10,000	10,000	10,000	10,000	10,000

Read the stated reeving first.

Using the wrong parts-of-line section gives the wrong gross capacity.

SOLUTION — PART 2 QUIZ QUESTION 1

PARTS OF LINE

SOLUTION PATH

1. **Read the stated reeving**
   4-part line
2. **Use the correct chart section**
   4-part chart
3. **Find the jib row**
   55 m jib
4. **Find the radius column**
   15 m

ANSWER

20,000 kg

Do not use the 2-part section.

COMEDIL CTT 561-20 — CAPACITY EXCERPTS

Jib configurations are rows. Hook radius values are columns. Capacities are kg.

COMEDIL CTT 561-20	4-PART LINE — capacity kg					
Radius (m)	5	10	15	20	25	30
55 m jib	20,000	20,000	20,000	20,000	20,000	20,000

COMEDIL CTT 561-20	2-PART LINE — distractor					
Radius (m)	5	10	15	20	25	30
55 m jib	10,000	10,000	10,000	10,000	10,000	10,000

Because the question states 4-part, read the 4-part section only.

55 m jib row + 15 m radius column = 20,000 kg gross capacity.

Fulford-style solution: choose reeving chart first, then jib row, then radius column.

FULFORD-STYLE
MULTI-STEP QUESTION

Crane:	**Comedil CTT 561-20**
Jib:	**85 m**
Hook radius:	**55 m**
Parts of line:	**2-part**
Load:	**3,900 kg**
Spreader bar:	**300 kg**
Slings / shackles:	**180 kg**

QUESTION

Can this load be lifted at 55 m radius?

If YES, what is the **remaining capacity?**

REFERENCE CHART EXCERPTS

Both parts-of-line sections are shown. Capacities are kg.

COMEDIL CTT 561-20	2-PART LINE — capacity kg				
Radius (m)	**45**	**50**	**55**	**60**	**65**
85 m jib	5,650	4,980	4,430	3,970	3,580

COMEDIL CTT 561-20	4-PART LINE — capacity kg				
Radius (m)	**45**	**50**	**55**	**60**	**65**
85 m jib	4,700	4,030	3,480	3,020	2,640

SELECT ONE:

- **A** Yes — 50 kg remaining
- **B** Yes — 350 kg remaining
- **C** No — gross load exceeds capacity
- **D** Yes — 530 kg remaining

Remember: everything hanging below the hook contributes to gross load.

SOLUTION — PART 2 QUIZ QUESTION 2

GROSS LOAD + CHART

SOLUTION PATH

1. **Calculate gross load**
   3,900 + 300 + 180 = 4,380 kg
2. **Use stated reeving**
   2-part line
3. **Read the chart**
   85 m jib row + 55 m radius
4. **Compare**
   4,380 kg ≤ 4,430 kg

ANSWER

YES — 50 kg remaining

Do not use the 4-part chart.

REFERENCE CHART EXCERPTS

Use the 2-part section because the question states 2-part line.

COMEDIL CTT 561-20	2-PART LINE — capacity kg				
Radius (m)	**45**	**50**	**55**	**60**	**65**
85 m jib	5,650	4,980	**4,430**	3,970	3,580

COMEDIL CTT 561-20	4-PART LINE — distractor				
Radius (m)	**45**	**50**	**55**	**60**	**65**
85 m jib	4,700	4,030	3,480	3,020	2,640

Gross load: 4,380 kg
Chart capacity: 4,430 kg
Remaining capacity: 4,430 − 4,380 = 50 kg

Correct choice: A — Yes, 50 kg remaining.

Fulford-style solution: gross load first, then correct reeving chart, then jib row and radius column.

LIFT CONFIGURATION

Comedil CTT 561-20 • 4-part line • 70 m jib installed

Load 5,400 kg + Spreader bar 300 kg + Slings / shackles 140 kg

COMEDIL CTT 561-20 — 4-PART LINE — GROSS CAPACITY (kg)

Jib / Radius	45	50	55	60	65	70	75	80	85
85 m	4,700	4,030	3,480	3,020	2,640	2,310	2,030	1,780	1,600
80 m	6,280	5,460	4,800	4,250	3,780	3,390	3,040	2,740	
75 m	7,920	6,940	6,140	5,480	4,920	4,440	4,100		
70 m	9,470	8,350	7,440	6,680	6,040	5,600			
65 m	10,240	9,040	8,070	7,260	6,700				
60 m	11,220	9,940	8,900	8,200					
55 m	12,790	11,360	10,400						

REMEMBER

Gross load includes the load and all stated rigging/accessory weights.

To find maximum radius, move outward until the next chart radius can no longer support the complete gross load.

DETERMINE

A. Which jib configuration applies?

B. What is the complete GROSS LOAD?

C. What is the maximum hook radius for this complete suspended load?

D. At that radius, how much capacity remains after load + rigging?

SELECT THE COMPLETE ANSWER:

- **A** 70 m jib | 65 m radius | 200 kg remains
- **B** 70 m jib | 70 m radius | 240 kg remains
- **C** 75 m jib | 60 m radius | 360 kg remains
- **D** 70 m jib | 60 m radius | 840 kg remains

Practice chart structure and values based on Fulford Certification's Comedil CTT 561-20 practice chart.

SOLUTION — PART 2 REPLACEMENT QUESTION

4-PART LINE

SOLUTION PATH

1. **CONFIGURATION**
   70 m jib / 4-part

2. **GROSS LOAD**
   5,400 + 300 + 140
   = 5,840 kg

3. **65 m RADIUS**
   Capacity = 6,040 kg
   PASS

4. **70 m RADIUS**
   Capacity = 5,600 kg
   FAIL

MAXIMUM RADIUS

65 m

70 m is 240 kg over capacity.

6,040 − 5,840 = 200 kg remains

COMEDIL CTT 561-20 — 4-PART GROSS CAPACITY (kg)

JIB CONFIGURATION / HOOK RADIUS									
Jib / Radius	45	50	55	60	65	70	75	80	85
85 m	4,700	4,030	3,480	3,020	2,640	2,310	2,030	1,780	1,600
80 m	6,280	5,460	4,800	4,250	3,780	3,390	3,040	2,800	
75 m	7,920	6,940	6,140	5,480	4,920	4,440	4,100		
70 m	9,470	8,350	7,440	6,680	6,040	5,600			
65 m	10,240	9,040	8,070	7,260	6,700				
60 m	11,220	9,940	8,900	8,200					
55 m	12,790	11,360	10,400						

WORK OUTWARD ON THE 70 m JIB ROW

Gross suspended load = 5,840 kg

65 m: 6,040 kg ≥ 5,840 kg → WITHIN CAPACITY

70 m: 5,600 kg < 5,840 kg → 240 kg OVER

CORRECT ANSWER

70 m jib | 65 m maximum radius | 200 kg remaining capacity

Remaining capacity includes the load, spreader bar, slings and shackles.

Chart values verified against Fulford Certification's Comedil CTT 561-20 practice chart.

WHAT DOES A HOIST GEAR CHANGE?

For this Comedil hoist: higher gear number = faster line speed, lower hoist capacity.

GEAR	2-PART CAPACITY	2-PART SPEED	4-PART CAPACITY	4-PART SPEED
1	10,000 kg	46 m/min	20,000 kg	23 m/min
2	7,140 kg	63 m/min	14,280 kg	31.5 m/min
3	5,000 kg	86 m/min	10,000 kg	43 m/min
4	3,460 kg	117 m/min	6,920 kg	58.5 m/min
5	600 kg	131 m/min	1,200 kg	65.5 m/min

THE PATTERN

GEAR 1 SLOWEST
HIGHEST HOIST CAPACITY

Speed increases
Capacity decreases

GEAR 5 FASTEST
LOWEST HOIST CAPACITY

CRITICAL DISTINCTION

HOIST-GEAR CAPACITY is not the same thing as CRANE LOAD-CHART CAPACITY.

Later, both limits must be checked. The lower applicable capacity controls the lift.

FIND THE HIGHEST USABLE GEAR

Start with the complete gross load — then choose the fastest gear that can carry it.

EXAMPLE

2-part line

Load	**10,700 lb**
Rigging	**+ 400 lb**
GROSS LOAD	**11,100 lb**

Question:

What is the highest usable hoist gear?

GEAR	2-PART MAX	SPEED	11,100 lb?
5	1,320 lb	430 fpm	NO
4	7,630 lb	384 fpm	NO
3	11,025 lb	282 fpm	NO
2	15,740 lb	206 fpm	YES
1	22,045 lb	150 fpm	YES

WORK FROM FASTEST DOWN

Gear 5	NO
Gear 4	NO
Gear 3	NO
Gear 2	YES

ANSWER: GEAR 2

Gear 3 is only rated for 11,025 lb — the 11,100 lb gross load is 75 lb too heavy.

Do not choose Gear 1 just because it has the greatest capacity. The question asks for the HIGHEST usable gear.

Teaching example based on the Fulford Comedil 110 AWL 100 LLC hoist speed/capacity table.

TWO LIMITS — WHICH ONE CONTROLS?

The load must be within BOTH the crane load chart and the selected hoist gear.

4-part line • 70 m jib • 60 m radius • Gross load = 6,500 kg • Selected gear = Gear 4

1 CRANE LOAD CHART

70 m JIB	50	55	60	65	70
Capacity	8,350	7,440	6,680	6,040	5,600

At 60 m radius: **6,680 kg**

2 HOIST GEAR CHART

GEAR	4-PART MAX kg	SPEED m/min
1	20,000	23
2	14,280	31.5
3	10,000	43
4	6,920	58.5

COMPARE THE TWO LIMITS

LOAD CHART **6,680 kg** vs. **GEAR 4** **6,920 kg** **LOWER LIMIT** **= 6,680 kg**

The 6,500 kg gross load passes BOTH limits.

But at this radius, the CRANE LOAD CHART controls because 6,680 kg is lower than Gear 4's 6,920 kg.

Key rule: when two limits apply, the lower applicable capacity governs.

WHEN THE HOIST GEAR BECOMES THE LIMIT

The crane chart may allow more — but the selected gear can still restrict the lift.

4-part line • 70 m jib • 55 m radius • Selected gear = Gear 4

CRANE LOAD CHART

70 m JIB	50	55	60	65	70
Capacity	8,350	7,440	6,680	6,040	5,600

70 m jib / 55 m radius = **7,440 kg**

HOIST GEAR CHART

GEAR	4-PART MAX kg	SPEED m/min
1	20,000	23
2	14,280	31.5
3	10,000	43
4	6,920	58.5

WHICH LIMIT GOVERNS?

Load chart **7,440 kg** **vs.** **Gear 4** **6,920 kg** **LOWER LIMIT =** **6,920 kg**

The crane structure allows 7,440 kg at this radius.

But in Gear 4, the hoist is limited to 6,920 kg — so GEAR 4 controls the allowable gross load.

Same rule every time: compare every applicable limit and use the lowest one.

SELECT THE HIGHEST USABLE HOIST GEAR

Fulford-style: calculate gross load first, then use the correct parts-of-line gear table.

QUESTION

Crane:	**Comedil CTT 561-20**
Hoist:	**110 AWL 100 LLC**
Parts of line:	**2-part**
Load weight:	**10,677 lb**
Rigging weight:	**419 lb**

What is the HIGHEST USABLE hoist gear?

Do not choose by speed alone.

HOIST SPEED & CAPACITY TABLE

Both 2-part and 4-part sections are shown. Capacity is in lb.

GEAR	2-PART CAPACITY	2-PART SPEED	4-PART CAPACITY	4-PART SPEED
1	22,045	150 fpm	44,090	75 fpm
2	15,740	206 fpm	31,480	103 fpm
3	11,025	282 fpm	22,045	141 fpm
4	7,630	384 fpm	15,260	192 fpm
5	1,320	430 fpm	2,640	215 fpm

SELECT ONE:

- **A** Gear 3
- **B** Gear 2
- **C** Gear 1
- **D** Gear 4

TRAP

Highest usable gear means fastest gear that can carry gross load.

SOLUTION — PART 3 PRACTICE QUESTION 1

HIGHEST USABLE GEAR

Given: 2-part line | Load 10,677 lb | Rigging 419 lb
Question: What is the highest usable hoist gear?

STEP 1

Calculate gross load

Load	**10,677 lb**
Rigging	**+ 419 lb**
Gross load	**11,096 lb**

STEP 2

Use the 2-part chart

Work down from the fastest gear until the **gross load fits.**

HOIST SPEED & CAPACITY TABLE

GEAR	2-PART CAPACITY	2-PART SPEED	RESULT
5	1,320 lb	430 fpm	NO
4	7,630 lb	384 fpm	NO
3	11,025 lb	282 fpm	NO
2	**15,740 lb**	**206 fpm**	**YES**
1	22,045 lb	150 fpm	YES

FULFORD-STYLE TRAP

Gear 3 looks close, but the gross load **is slightly over its rating.**

Choose the fastest gear that still fits.

WHY GEAR 2?

Gross load **11,096 lb**

Gear 3
11,025 lb
Short by 71 lb

Gear 2
15,740 lb
Can carry the gross load

ANSWER
Gear 2

Method: gross load first → correct parts-of-line table → fastest gear that can carry the gross load.

DETERMINE IF THE LIFT CAN BE MADE IN GEAR 4

Comedil CTT 561-20

Determine if the lift can be **performed using Gear 4.**

- 4 parts of line
- Jib length – 70 metres
- Hook radius – 55 metres
- Load weight – 6,500 kg
- Rigging weight – 300 kg

Answer:

YES / NO

Show your calculation.

4-PART LOAD CHART

Jib / R	45	50	55	60	65	70
75 m	7,920	6,940	6,140	5,480	4,920	4,440
70 m	9,470	8,350	7,440	6,680	6,040	5,600
65 m	10,240	9,040	8,070	7,260	6,700	

HOIST SPEED & CAPACITY

GEAR	2-PART kg	2-PART m/min	4-PART kg	4-PART m/min
1	10,000	46	20,000	23
2	7,140	63	14,280	31.5
3	5,000	86	10,000	43
4	3,460	117	6,920	58.5
5	600	131	1,200	65.5

Use the supplied charts. Do not assume the crane load chart is the only limit.

Training question based on Fulford's Comedil CTT 561-20 practice chart and 110 AWL 100 LLC hoist table.

SOLUTION — PART 3 PRACTICE QUESTION 2

4-part • 70 m jib • 55 m radius • Gear 4 • Load 6,500 kg • Rigging 300 kg

1 GROSS LOAD

6,500 + 300

= 6,800 kg

2 CHECK BOTH LIMITS

Crane chart **7,440 kg**

Gear 4 **6,920 kg**

LOWER LIMIT

6,920 kg

Gear 4 controls

6,920 − 6,800 = **120 kg**

4-PART LOAD CHART

70 m JIB	45	50	55	60	65	70
Capacity	9,470	8,350	7,440	6,680	6,040	5,60

HOIST SPEED & CAPACITY

GEAR	2-PART kg	4-PART kg
1	10,000	20,000
2	7,140	14,280
3	5,000	10,000
4	3,460	6,920
5	600	1,200

RESULT

Crane chart:

7,440 kg

Gear 4:

6,920 kg

Gross load:

6,800 kg

YES the lift can be made in Gear 4.

Gear 4 controls • 120 kg remains

Trap: the crane chart leaves 640 kg, but the selected hoist gear leaves only 120 kg.

CAN THIS LIFT BE MADE IN GEAR 3?

Comedil CTT 561-20

Determine if the lift can be **performed using Gear 3.**

- 4 parts of line
- Jib length – 70 metres
- Hook radius – 60 metres
- Load weight – 6,600 kg
- Rigging weight – 200 kg

Answer:

YES / NO

Show your calculation.

4-PART LOAD CHART

Jib / R	45	50	55	60	65	70
75 m	7,920	6,940	6,140	5,480	4,920	4,440
70 m	9,470	8,350	7,440	6,680	6,040	5,600
65 m	10,240	9,040	8,070	7,260	6,700	

HOIST SPEED & CAPACITY

GEAR	2-PART kg	2-PART m/min	4-PART kg	4-PART m/min
1	10,000	46	20,000	23
2	7,140	63	14,280	31.5
3	5,000	86	10,000	43
4	3,460	117	6,920	58.5
5	600	131	1,200	65.5

Use the supplied charts and all weights given in the question.

SOLUTION — PART 3 PRACTICE QUESTION 3

CRANE CHART FAILS

4-part • 70 m jib • 60 m radius • Gear 3 • Load 6,600 kg • Rigging 200 kg

SOLUTION PATH

1 Gross load

6,600 + 200

= 6,800 kg

2 Crane load chart

70 m jib / 60 m radius

= 6,680 kg

FAIL — 120 kg over

3 Hoist gear

Gear 3 / 4-part

= 10,000 kg

PASS

VERIFY THE TWO LIMITS

4-PART LOAD CHART

70 m JIB	50	55	60	65	70
Capacity	8,350	7,440	6,680	6,040	5,600

HOIST GEAR TABLE

GEAR	4-PART kg	m/min
1	20,000	23
2	14,280	31.5
3	10,000	43
4	6,920	58.5
5	1,200	65.5

RESULT

Gross load **6,800 kg**

Gear 3 limit **10,000 kg**

PASS

Crane chart limit **6,680 kg**

FAIL

6,800 − 6,680

= 120 kg OVER

ANSWER: NO

The crane chart is exceeded.

Trap: Gear 3 has enough hoist capacity, but the 70 m jib / 60 m radius crane chart is too low.

PART 4 POWER — READING THE NUMBERS

WHAT DO THE POWER TERMS MEAN?

Recognize the terms you may see on a crane specification or hoist chart.

V / kV VOLTS / KILOVOLTS

Electrical voltage

480 V

kV = 1,000 volts

A AMPERES

Electrical current

300 A

Amount of current

PH PHASE

Type of AC supply

3-PHASE

Common crane supply

Hz HERTZ

Electrical frequency

60 Hz

Cycles per second

kVA KILOVOLT-AMPS

Supply / apparent power

216 kVA

Electrical supply rating

hp / kW MOTOR POWER

Hoist motor output

147 hp / 110 kW

Two units of motor power

FULFORD COMEDIL EXAMPLE: 480 V • 3-PHASE • 60 Hz • 300 A • 216 kVA | Hoist: 147 hp / 110 kW

Know what each term describes. No electrical calculations are required for this lesson.

CHECK THE INSTALLED HOIST UNIT FIRST

The same crane model may be equipped with a different hoist unit.

FULFORD / COMEDIL EXAMPLE

Hoist Unit

110 AWL 100 LLC

HOIST POWER

147 hp / 110 kW

POWER REQUIREMENTS

480 V • 3-PHASE • 60 Hz

300 A SERVICE • 216 kVA

BEFORE USING THE HOIST TABLE

1. **Identify the installed hoist unit**
2. **Match it to the correct chart**
3. **Use that unit's capacity & speed data**

Different hoist unit = different hoist data

KEY POINT **Do not assume the hoist unit shown on a sample chart is the unit installed on the crane.**

Training slide based on Fulford's Comedil CTT 561-20 practice chart.

PART 4 — POWER OPTIONS

THE POWER OPTION CHANGES THE HOIST TABLE

Read the installed hoist power first — then use the matching speed and capacity data.

INSTALLED HOIST

110 AWL 100 LLC

147 hp / 110 kW

THIS IDENTIFIES

which hoist table belongs to the crane.

Do not mix hoist options.

MATCHING HOIST SPEED & CAPACITY

110 AWL 100 LLC — 147 hp (110 kW)

GEAR	2-PART CAP.	2-PART SPEED	4-PART CAP.	4-PART SPEED
1	**10,000 kg**	**46 m/min**	**20,000 kg**	**23 m/min**
2	7,140 kg	63 m/min	14,280 kg	31.5 m/min
3	5,000 kg	86 m/min	10,000 kg	43 m/min
4	3,460 kg	117 m/min	6,920 kg	58.5 m/min
5	600 kg	131 m/min	1,200 kg	65.5 m/min

KEY POINT Power / hoist option first → matching table second → then read gear capacity and speed.

PART 4 — POWER OPTIONS

SAME CRANE — TWO HOIST POWER OPTIONS

Liebherr 200 EC-H 12 example

45 kW HOIST

Power requirement — upper crane

61 kW / 68 kVA

GEAR	CAPACITY	LINE SPEED
Gear 1	**12,000 kg**	**15 m/min**
Gear 2	6,200 kg	30 m/min
Gear 4	2,900 kg	61 m/min

Different power option → different hoist performance

65 kW HOIST

Power requirement — upper crane

81 kW / 88 kVA

GEAR	CAPACITY	LINE SPEED
Gear 1	**12,000 kg**	**20 m/min**
Gear 3	5,700 kg	48 m/min
Gear 7	2,200 kg	112 m/min

Different power option → different hoist performance

KEY POINT **The crane model is the same, but the installed hoist power changes the applicable hoist data. Identify the hoist option before answering capacity or line-speed questions.**

Source example: Liebherr 200 EC-H 12 technical data.

PART 4 — POWER OPTIONS

kW vs kVA — DON'T MIX THEM UP

They describe two different things on the crane specification.

kW — HOIST MOTOR

What it tells you:

Which hoist / winch power option is installed.

Liebherr example

45 kW HOIST

65 kW HOIST

kVA — CONNECTED LOAD

What it tells you:

Electrical supply / connected load requirement.

Liebherr example

45 kW HOIST → 63 kVA

65 kW HOIST → 86 kVA

COMMON TRAP

Crane has a 65 kW hoist. **Question asks for connected load.** **Answer: 86 kVA**

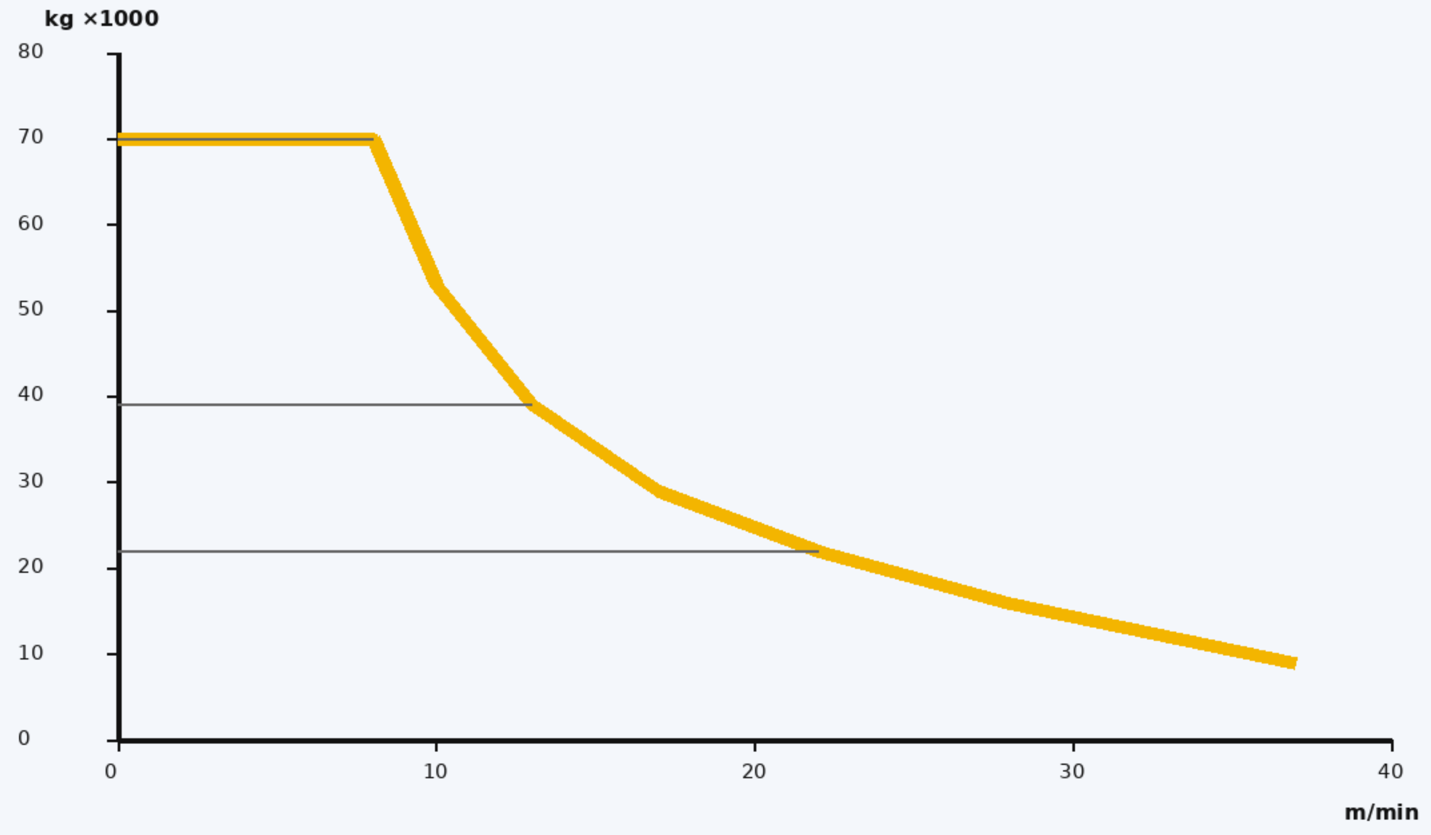
MEMORY RULE: **kW = HOIST MOTOR • kVA = ELECTRICAL / CONNECTED LOAD**

Liebherr 200 EC-H 12 technical data example: 45 kW / 63 kVA and 65 kW / 86 kVA.

PART 4 — PRACTICE QUESTION

READ THE HOIST PERFORMANCE GRAPH

Liebherr 630 EC-H 70 Litronic • 110 kW FU hoist



CRANE DATA

Hoist unit

110 kW FU

Connected load

153 kVA

QUESTION

A load of 22,000 kg is on the hook.

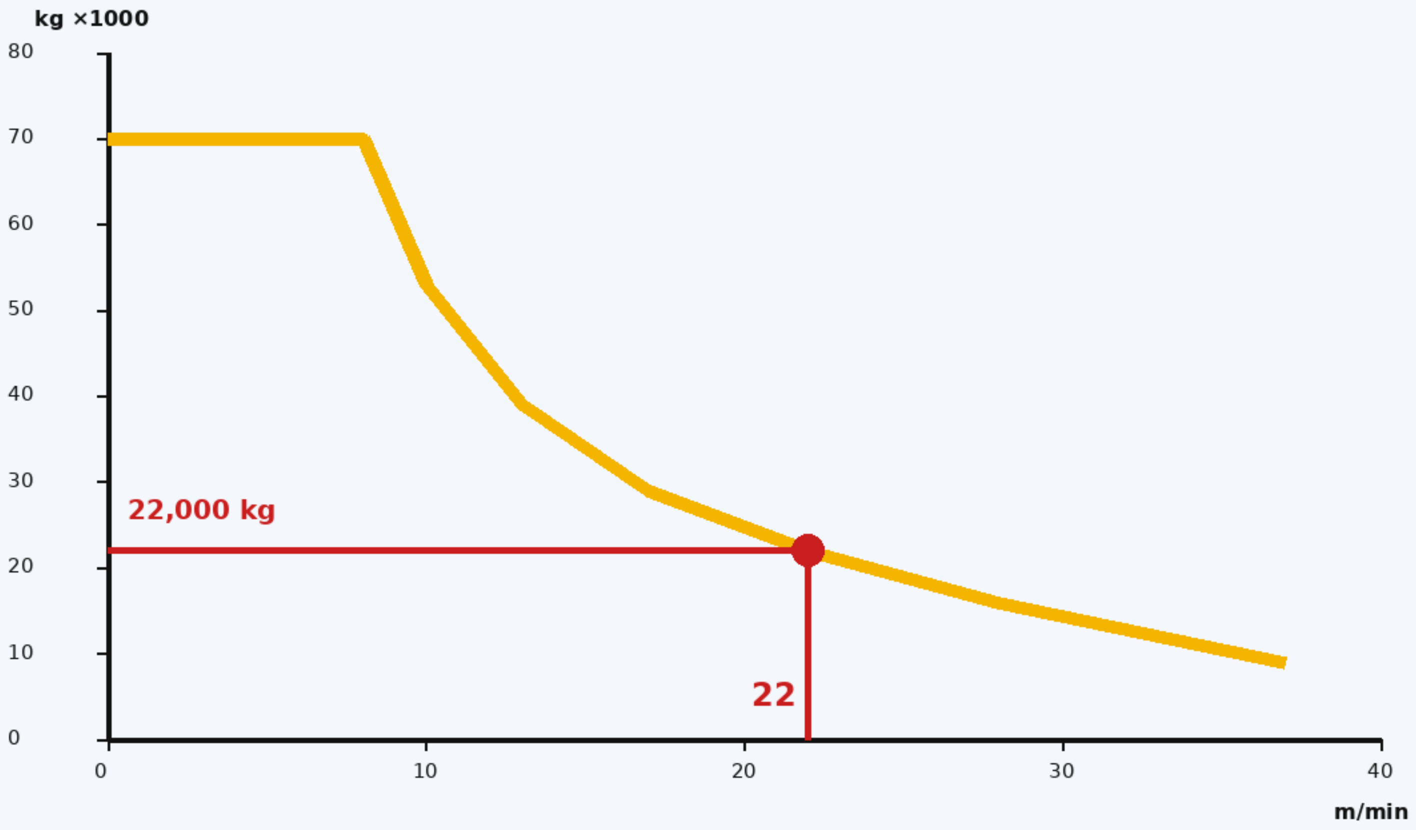
Using the hoist performance graph, determine the **MAXIMUM hoisting** speed available.

Read the graph — do not calculate electrical power.

PART 4 — SOLUTION

READ THE HOIST PERFORMANCE GRAPH

Liebherr 630 EC-H 70 Litronic • 110 kW FU hoist



SOLUTION

1 **Find 22,000 kg**
on the load axis.

2 **Move across**
to the performance curve.

3 **Move down to**
the speed axis.

MAXIMUM SPEED

22 m/min

22,000 kg → performance curve → 22 m/min

PART 5

ADVANCED CHART READING

FULFORD-STYLE CHART SKILLS & TRAPS

JIB CONFIGURATION

Compare the correct installed jib

TEST WEIGHTS

Read capacity, then apply the test %

LOAD-LINE WEIGHT

Know what rope weight must be accounted for

TROLLEY SPEED

Recognize when the question is not capacity

MIXED CHARTS

Choose the chart that actually answers the question

THE GOAL

Don't just read a number — identify the configuration, select the right chart, account for the deductions, and answer exactly what the question asks.

PART 5 — ADVANCED CHART READING

SAME RADIUS — DIFFERENT JIB LENGTH

Liebherr 200 HC — Fulford-style radius and capacity chart

JIB LENGTH	50 ft 15.2 m	60 ft 18.3 m	70 ft 21.3 m	80 ft 24.4 m	90 ft 27.4 m	100 ft 30.5 m	110 ft 33.5 m	120 ft 36.6 m
197 ft / 60.0 m	—	—	—	—	13,120 5,950	—	—	—
180 ft / 54.9 m	—	—	—	—	14,550 6,600	—	—	—
164 ft / 50.0 m	—	—	—	—	15,980 7,250	—	—	—
148 ft / 45.1 m	—	—	—	—	17,420 7,900	—	—	—
131 ft / 39.9 m	—	—	—	—	18,960 8,600	—	—	—
104 ft / 31.7 m	—	—	—	—	20,930 9,490	—	—	—

READ THE SAME RADIUS COLUMN

60.0 m jib → **5,950 kg**　　31.7 m jib → **9,490 kg**

CAPACITY DIFFERENCE

9,490 − 5,950 = 3,540 kg

LESSON: Find the radius column first, then read the capacity from each applicable jib-length row.

PART 5 — ADVANCED CHART READING

TIP TRIP TEST BLOCK — 5% OF GROSS CAPACITY

Comedil CTT 561-20 — 2-PART LINE — Fulford Practice Question 4

JIB LENGTH	45 m RADIUS	50 m RADIUS	55 m RADIUS	60 m RADIUS	65 m RADIUS	70 m RADIUS
85 m	5,650 kg	4,980 kg	4,430 kg	3,970 kg	3,580 kg	3,250 kg
80 m	7,160 kg	6,340 kg	5,670 kg	5,120 kg	4,650 kg	4,250 kg
75 m	8,670 kg	7,690 kg	6,900 kg	6,240 kg	5,690 kg	5,210 kg
70 m	10,000 kg	9,040 kg	8,140 kg	7,380 kg	6,740 kg	6,200 kg
65 m	10,000 kg	9,770 kg	8,790 kg	7,980 kg	7,300 kg	
60 m	10,000 kg	10,000 kg	9,580 kg	8,700 kg		

STEP 1 — READ THE 2-PART CHART

70 m jib × 70 m radius = 6,200 kg gross capacity

STEP 2 **6,200 kg × 5% = 6,200 × 0.05 = 310 kg**

IMPORTANT

Use the 2-PART chart.
Not the 4-part chart.

TIP TRIP TEST BLOCK = 310 kg

PART 5 — ADVANCED CHART READING

HOOK TRAVEL — CAPACITY REDUCTION

FULFORD PRACTICE QUESTION 7 — WILBERT WT 200 e.tronic

65 m jib • 57.5 m hook radius • Hook lowered 100 m

Determine the capacity reduction for the additional hoist rope.

JIB	50.0 m	52.5 m	55.0 m	57.5 m	60.0 m	62.5 m	65.0 m
65.0 m	2.4 t	2.2 t	2.1 t	**2.0 t**	1.9 t	1.8 t	1.7 t
62.5 m	2.9 t	2.8 t	2.6 t	2.5 t	2.4 t	2.3 t	
60.0 m	3.2 t	3.0 t	2.8 t	2.7 t	2.6 t		
57.5 m	3.5 t	3.3 t	3.1 t	3.0 t			

WILBERT CHART NOTE

40 m hook travel included • double rope / 2-part = 2.6 kg/m

FULFORD SOLUTION

1. Read capacity

65 m jib × 57.5 m radius = 2.0 t

2. Extra hook travel

100 m − 40 m = 60 m

3. Rope reduction

60 m × 2.6 kg/m = 156 kg

ANSWER

Capacity reduction = 156 kg

The extra 60 m of suspended 2-part hoist rope is the deduction Fulford is asking for.

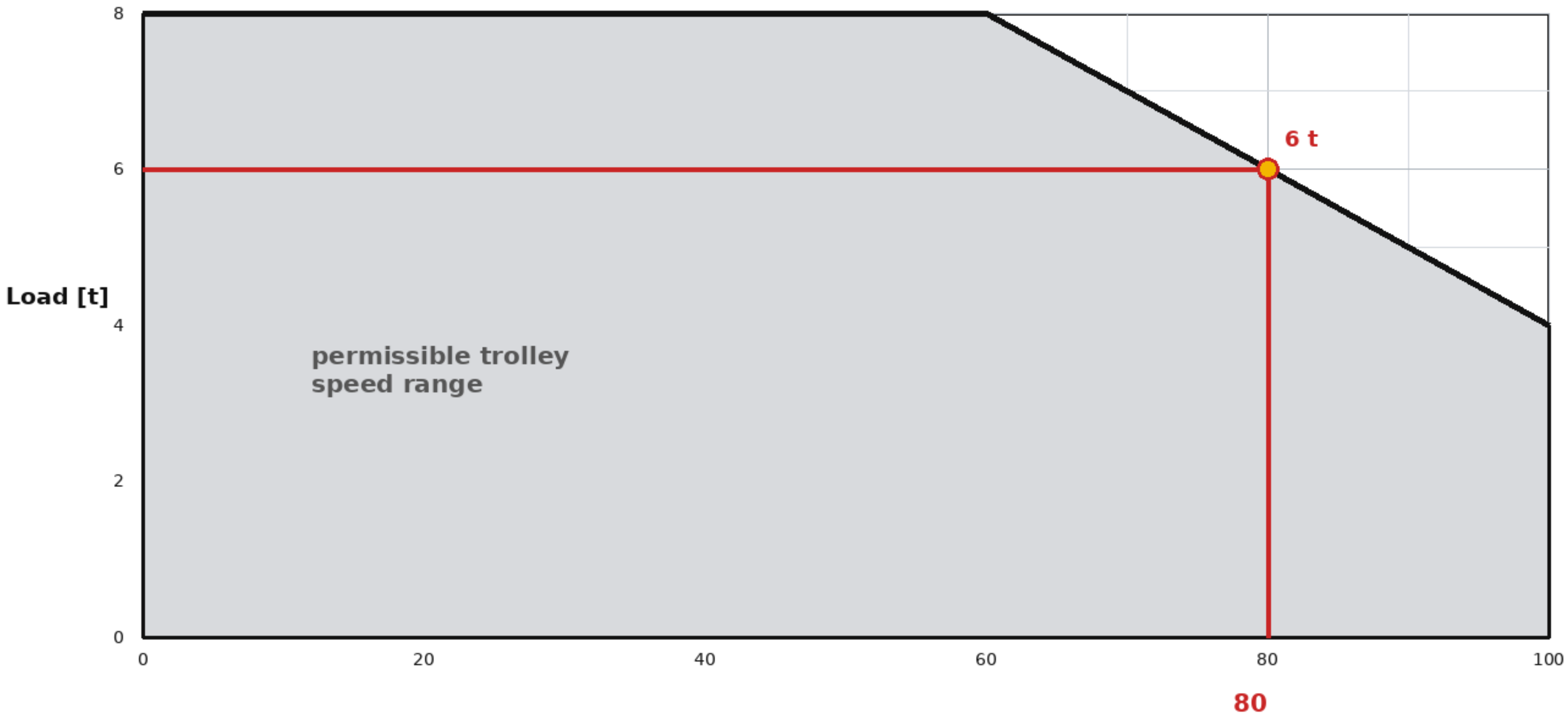
PART 5 — ADVANCED CHART READING

TROLLEY SPEED — WILBERT PERFORMANCE GRAPH

FULFORD PRACTICE QUESTION 8 — WILBERT WT 200 e.tronic

Load weight: 6 tonnes **Determine the maximum trolley speed.**

TW — MOVING TROLLEY 7.5 kW



HOW TO READ IT

1. Find 6 t on the load axis.
2. Move across to the sloped envelope line.
3. Drop down to the trolley-speed axis.

80 m/min MAX

FULFORD ANSWER: **6 tonnes → maximum trolley speed = 80 m/min**

PART 5 — ADVANCED CHART READING

EXAM SKILL — WHAT INFORMATION ACTUALLY MATTERS?

FULFORD-STYLE QUESTION — WILBERT WT 200 e.tronic

65 m jib | 57.5 m radius | **6 tonne load** | 100 m hook travel

Question: What is the maximum trolley speed?

STEP 1 — READ THE QUESTION FIRST

The question asks for:

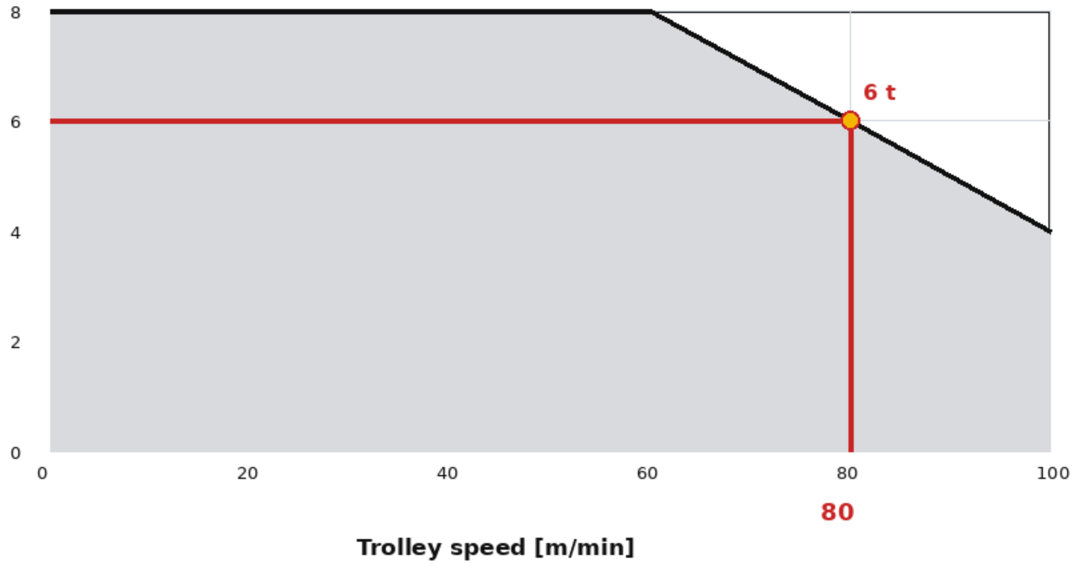
TROLLEY SPEED

Needed — **6 tonne load**

Not needed — 65 m jib
57.5 m radius
100 m hook travel

Do not solve information the question never asked for.

STEP 2 — GO TO THE CORRECT CHART



ANSWER: 80 m/min **The load is what you need to read the trolley-speed graph.**

EXAM HABIT: Identify what is being asked → choose the chart → ignore information that does not affect that answer.

PART 5 — QUIZ

QUESTION 1 — FIND IT IN THE REAL LOAD CHART

LIEBHERR 200 HC — 2-PART OPERATION

At a 27.4 m (90 ft) radius, compare the 60.0 m jib with the 31.7 m jib.
What is the difference in gross lifting capacity?

JIB (m)	18.3 m	21.3 m	24.4 m	27.4 m	30.0 m	31.7 m	33.5 m	36.7 m
60.0 m	8,000	7,495	6,710	5,950	5,150	4,860	4,600	4,140
55.0 m	9,420	8,170	7,160	6,430	5,670	5,350	5,030	4,560
48.3 m	10,000	9,070	7,970	7,100	6,320	5,960	5,580	5,100
43.3 m	10,000	9,960	8,820	7,860	7,020	6,640	6,190	5,600
36.7 m	10,000	10,000	9,810	8,800	7,880	7,450	6,930	6,400
31.7 m	10,000	10,000	10,000	9,490	8,510	8,050		

Gross capacities shown in kg. Values reproduced from the Liebherr 200 HC load chart used in Fulford practice material.

- **A** 2,540 kg
- **B** 3,540 kg
- **C** 3,950 kg
- **D** 4,540 kg

PART 5 — QUIZ SOLUTION

QUESTION 1 — JIB CONFIGURATION

LIEBHERR 200 HC — 2-PART OPERATION

At 27.4 m radius, compare the 60.0 m jib with the 31.7 m jib.
Find both chart capacities, then subtract.

JIB (m)	18.3 m	21.3 m	24.4 m	27.4 m	30.0 m	31.7 m	33.5 m	36.7 m
60.0 m	8,000	7,495	6,710	**5,950**	5,150	4,860	4,600	4,140
55.0 m	9,420	8,170	7,160	6,430	5,670	5,350	5,030	4,560
48.3 m	10,000	9,070	7,970	7,100	6,320	5,960	5,580	5,100
43.3 m	10,000	9,960	8,820	7,860	7,020	6,640	6,190	5,600
36.7 m	10,000	10,000	9,810	8,800	7,880	7,450	6,930	6,400
31.7 m	10,000	10,000	10,000	**9,490**	8,510	8,050		

SOLUTION

60.0 m jib at 27.4 m radius = 5,950 kg

31.7 m jib at 27.4 m radius = 9,490 kg

Difference: 9,490 − 5,950 = 3,540 kg

CORRECT ANSWER

B — 3,540 kg

Method: same radius column, two different jib rows, then subtract the lower capacity from the higher capacity.

PART 5 — QUIZ

QUESTION 2 — TIP TEST BLOCK

COMEDIL CTT 561-20 — 2-PART LINE

The crane is configured with a 75 m jib. A 5% tip-trip test block is required.
Using the capacity at the maximum radius for this configuration, determine the test-block weight.

JIB	45 m	50 m	55 m	60 m	65 m	70 m	75 m
85 m	5,650	4,980	4,430	3,970	3,580	3,250	2,970
80 m	7,160	6,340	5,670	5,120	4,650	4,250	3,900
75 m	8,670	7,690	6,900	6,240	5,690	5,210	4,800
70 m	10,000	9,040	8,140	7,380	6,740	6,200	
65 m	10,000	9,770	8,790	7,980	7,300		
60 m	10,000	10,000	9,580	8,700			

Gross capacities shown in kg. First identify the maximum radius available for the 75 m jib.

A 240 kg

B 260.5 kg

C 480 kg

D 4,800 kg

PART 5 — QUIZ SOLUTION

QUESTION 2 — TIP TEST BLOCK

COMEDIL CTT 561-20 — 2-PART LINE

75 m jib configuration • 5% tip-trip test block

Use the capacity at the maximum radius for this jib configuration.

JIB	45 m	50 m	55 m	60 m	65 m	70 m	75 m
85 m	5,650	4,980	4,430	3,970	3,580	3,250	2,970
80 m	7,160	6,340	5,670	5,120	4,650	4,250	3,900
75 m	8,670	7,690	6,900	6,240	5,690	5,210	**4,800**
70 m	10,000	9,040	8,140	7,380	6,740	6,200	
65 m	10,000	9,770	8,790	7,980	7,300		
60 m	10,000	10,000	9,580	8,700			

Step 1: The maximum radius for the 75 m jib is 75 m. Step 2: Read the tip capacity: 4,800 kg.

CALCULATION

Tip capacity: **4,800 kg**

5% test block: **4,800 × 0.05 = 240 kg**

CORRECT ANSWER

A — 240 kg

PART 5 — QUIZ

QUESTION 3 — MAXIMUM ALLOWABLE LOAD

WILBERT WT 200 e.tronic

65 m jib **57.5 m hook radius** **2-part / double rope** **Hook lowered 90 m**

What is the maximum allowable gross load after accounting for the additional hoist-rope weight?

JIB	50.0 m	52.5 m	55.0 m	57.5 m	60.0 m	62.5 m	65.0 m
65.0 m	2.4 t	2.2 t	2.1 t	2.0 t	1.9 t	1.8 t	1.7 t
62.5 m	2.9 t	2.8 t	2.6 t	2.5 t	2.4 t	2.3 t	
60.0 m	3.2 t	3.0 t	2.8 t	2.7 t	2.6 t		
57.5 m	3.5 t	3.3 t	3.1 t	3.0 t			
55.0 m	3.8 t	3.6 t	3.4 t				

CHART NOTE

Rated capacities include 40 m hook travel. For double rope / 2-part operation, deduct 2.6 kg for every metre of hook travel over 40 m.

A 1,870 kg

B 1,740 kg

C 2,000 kg

D 2,130 kg

This question asks for the final allowable gross load — not just the rope-weight reduction.

PART 5 — QUIZ SOLUTION

QUESTION 3 — MAXIMUM ALLOWABLE LOAD

WILBERT WT 200 e.tronic

65 m jib • 57.5 m radius • 2-part / double rope • hook lowered 90 m

Question asks for final allowable gross load after rope-weight deduction.

JIB	50.0 m	52.5 m	55.0 m	57.5 m	60.0 m	62.5 m	65.0 m
65.0 m	2.4 t	2.2 t	2.1 t	2.0 t	1.9 t	1.8 t	1.7 t
62.5 m	2.9 t	2.8 t	2.6 t	2.5 t	2.4 t	2.3 t	
60.0 m	3.2 t	3.0 t	2.8 t	2.7 t	2.6 t		
57.5 m	3.5 t	3.3 t	3.1 t	3.0 t			
55.0 m	3.8 t	3.6 t	3.4 t				

CHART NOTE

40 m hook travel included. 2-part/double rope: deduct 2.6 kg per metre over 40 m.

SOLUTION PATH

1. **Chart capacity**
   2.0 t = 2,000 kg
2. **Extra hook travel**
   90 m − 40 m = 50 m
3. **Rope deduction**
   50 m × 2.6 kg/m = 130 kg
4. **Allowable gross load**
   2,000 − 130 = 1,870 kg

ANSWER

Maximum allowable gross load = 1,870 kg

Correct choice: A The 130 kg rope-weight deduction must be subtracted from the 2,000 kg chart capacity.

LIEBHERR

TOWER CRANE CHARTS INTRO

Liebherr 71 EC-B 5

Load Chart Fundamentals

Read
Understand Capacity

Select
Radius Configuration

Lift
Safely Within Limits

LIEBHERR 71 EC-B 5

QUESTION 1 – GROSS LOAD / CAPACITY (2-PART)

QUESTION 1

Parts of line:	2-part
Jib length:	50.0 m
Hook radius:	35.0 m
Load weight:	1,400 kg
Rigging weight:	125 kg

CALCULATION

Gross load

1,400 kg + 125 kg

= **1,525 kg**
(gross)

CHART CAPACITY

From 2-part chart:

50.0 m jib → 35.0 m radius

= **1,580 kg**
(rated capacity)

CAPACITY REMAINING

1,580 kg – 1,525 kg

= **55 kg** remaining

ANSWER: YES – *Lift is within rated capacity.*

2-PART CHART – MAXIMUM CAPACITY 2,500 kg

Ausladung und Tragfähigkeit Radius and capacity / Portée et charge / Sbraccio e portata / Alcances y cargas / Alcance e capacidade de carga

m r	m/kg	15,0	17,5	20,0	22,5	25,0	27,5	30,0	32,5	35,0	37,5	40,0	42,5	45,0	47,5	50,0
50,0 (r = 51,5)	2,4–23,7 **2500**	2500	2500	2500	2500	2350	2110	1900	1730	**1580**	1450	1340	1240	1150	1070	**1000**
47,5 (r = 49,0)	2,4–25,0 **2500**	2500	2500	2500	2500	2500	2240	2030	1840	1690	1550	1430	1330	1230	**1150**	
45,0 (r = 46,5)	2,4–26,1 **2500**	2500	2500	2500	2500	2500	2350	2130	1940	1770	1630	1510	1400	**1300**		
42,5 (r = 44,0)	2,4–26,9 **2500**	2500	2500	2500	2500	2500	2430	2200	2010	1840	1690	1560	**1450**			
40,0 (r = 41,5)	2,4–27,4 **2500**	2500	2500	2500	2500	2500	2490	2250	2050	1880	1730	**1600**				
37,5 (r = 39,0)	2,4–28,3 **2500**	2500	2500	2500	2500	2500	2500	2340	2130	1950	**1800**					
35,0 (r = 36,5)	2,4–28,9 **2500**	2500	2500	2500	2500	2500	2500	2390	2180	**2000**						
32,5 (r = 34,0)	2,4–29,7 **2500**	2500	2500	2500	2500	2500	2500	2470	**2250**							
30,0 (r = 31,5)	2,4–30,0 **2500**	2500	2500	2500	2500	2500	2500	**2500**								
27,5 (r = 29,0)	2,4–27,5 **2500**	2500	2500	2500	2500	2500	**2500**									
25,0 (r = 26,5)	2,4–25,0 **2500**	2500	2500	2500	2500	**2500**										
22,5 (r = 24,0)	2,4–22,5 **2500**	2500	2500	2500	**2500**											
20,0 (r = 21,5)	2,4–20,0 **2500**	2500	2500	**2500**												

NOTE: Use working hook-radius columns across the chart. Do **not** use *jib-end r* dimension.

QUESTION 2 — MAXIMUM RADIUS

Liebherr 71 EC-B 5 • 2-part line

LOAD	1,700 kg
+ RIGGING	90 kg
GROSS LOAD	**1,790 kg**

45.0 m JIB

Find the farthest radius that still carries 1,790 kg.

32.5 m	**1,940 kg ✓**
35.0 m	**1,770 kg ×**

MAXIMUM RADIUS

32.5 m

m r	m/kg	m/kg 15,0	17,5	20,0	22,5	25,0	27,5	30,0	32,5	35,0	37,5	40,0	42,5	45,0	47,5	50,0
50,0 (r = 51,5)	2,4–23,7 **2500**	2500	2500	2500	2500	2350	2110	1900	1730	1580	1450	1340	1240	1150	1070	**1000**
47,5 (r = 49,0)	2,4–25,0 **2500**	2500	2500	2500	2500	2500	2240	2030	1840	1690	1550	1430	1330	1230	**1150**	
45,0 (r = 46,5)	2,4–26,1 **2500**	2500	2500	2500	2500	2500	2350	2130	1940	1770	1630	1510	1400	**1300**		
42,5 (r = 44,0)	2,4–26,9 **2500**	2500	2500	2500	2500	2500	2430	2200	2010	1840	1690	1560	**1450**			
40,0 (r = 41,5)	2,4–27,4 **2500**	2500	2500	2500	2500	2500	2490	2250	2050	1880	1730	**1600**				
37,5 (r = 39,0)	2,4–28,3 **2500**	2500	2500	2500	2500	2500	2500	2340	2130	1950	**1800**					
35,0 (r = 36,5)	2,4–28,9 **2500**	2500	2500	2500	2500	2500	2500	2390	2180	**2000**						
32,5 (r = 34,0)	2,4–29,7 **2500**	2500	2500	2500	2500	2500	2500	2470	**2250**							
30,0 (r = 31,5)	2,4–30,0 **2500**	2500	2500	2500	2500	2500	2500	**2500**								
27,5 (r = 29,0)	2,4–27,5 **2500**	2500	2500	2500	2500	2500	**2500**									
25,0 (r = 26,5)	2,4–25,0 **2500**	2500	2500	2500	2500	**2500**										
22,5 (r = 24,0)	2,4–22,5 **2500**	2500	2500	2500	**2500**											
20,0 (r = 21,5)	2,4–20,0 **2500**	2500	2500	**2500**												

32.5 m is the last listed radius with capacity ≥ 1,790 kg.

QUESTION 3 — DIFFERENT JIB CONFIGURATIONS

Liebherr 71 EC-B 5 • 2-part line • Same radius, different jib length

HOOK RADIUS 35.0 m

50.0 m JIB 1,580 kg

40.0 m JIB 1,880 kg

DIFFERENCE

1,880 − 1,580

= 300 kg

ANSWER

300 kg more capacity

The shorter 40.0 m jib has
more capacity at the same radius.

m r	m/kg	m/kg 15,0	17,5	20,0	22,5	25,0	27,5	30,0	32,5	35,0	37,5	40,0	42,5	45,0	47,5	50,0
50,0 (r = 51,5)	2,4–23,7 **2500**	2500	2500	2500	2500	2350	2110	1900	1730	1580	1450	1340	1240	1150	1070	**1000**
47,5 (r = 49,0)	2,4–25,0 **2500**	2500	2500	2500	2500	2500	2240	2030	1840	1690	1550	1430	1330	1230	**1150**	
45,0 (r = 46,5)	2,4–26,1 **2500**	2500	2500	2500	2500	2500	2350	2130	1940	1770	1630	1510	1400	**1300**		
42,5 (r = 44,0)	2,4–26,9 **2500**	2500	2500	2500	2500	2500	2430	2200	2010	1840	1690	1560	**1450**			
40,0 (r = 41,5)	2,4–27,4 **2500**	2500	2500	2500	2500	2500	2490	2250	2050	1880	1730	**1600**				
37,5 (r = 39,0)	2,4–28,3 **2500**	2500	2500	2500	2500	2500	2500	2340	2130	1950	**1800**					
35,0 (r = 36,5)	2,4–28,9 **2500**	2500	2500	2500	2500	2500	2500	2390	2180	**2000**						
32,5 (r = 34,0)	2,4–29,7 **2500**	2500	2500	2500	2500	2500	2500	2470	**2250**							
30,0 (r = 31,5)	2,4–30,0 **2500**	2500	2500	2500	2500	2500	2500	**2500**								
27,5 (r = 29,0)	2,4–27,5 **2500**	2500	2500	2500	2500	2500	**2500**									
25,0 (r = 26,5)	2,4–25,0 **2500**	2500	2500	2500	2500	**2500**										
22,5 (r = 24,0)	2,4–22,5 **2500**	2500	2500	2500	**2500**											
20,0 (r = 21,5)	2,4–20,0 **2500**	2500	2500	**2500**												

At the same 35.0 m radius: 40.0 m jib capacity is higher than 50.0 m jib capacity.

LIEBHERR 71 EC-B 5 QUESTION 4 – AVAILABLE CAPACITY (4-PART)

QUESTION 4

Parts of line:	4-part
Jib length:	47.5 m
Hook radius:	40.0 m
Load weight:	1,100 kg
Rigging weight:	100 kg

Determine how much additional gross load capacity remains.

CALCULATIONS

Gross load

1,100 kg + 100 kg

= **1,200 kg** (gross)

CHART CAPACITY (4-PART)

From 4-part chart:

47.5 m jib → 40.0 m radius

= **1,290 kg**

(rated capacity)

AVAILABLE CAPACITY REMAINING

1,290 kg (rated) – **1,200 kg** (gross)

= **90 kg** remaining

ANSWER:

90 kg *of additional gross load capacity remains.*

4-PART CHART – MAXIMUM CAPACITY 5,000 kg

Ausladung und Tragfähigkeit

Radius and capacity / Portée et charge / Sbraccio e portata / Alcances y cargas / Alcance e capacidade de carga

m r	m/kg	+	m/kg 15,0	17,5	20,0	22,5	25,0	27,5	30,0	32,5	35,0	37,5	40,0	42,5	45,0	47,5	50,0
50,0 (r = 51,5)	2,4–22,9 2500	2,4–12,8 5000	4150	3470	2950	2560	2250	1990	1780	1600	1450	1310	1200	1090	1000	920	850
47,5 (r = 49,0)	2,4–24,1 2500	2,4–13,4 5000	4400	3680	3140	2730	2390	2120	1900	1710	1550	1410	1290	1180	1090	**1000**	
45,0 (r = 46,5)	2,4–25,1 2500	2,4–14,0 5000	4600	3850	3290	2860	2510	2230	2000	1800	1630	1490	1360	1250	1150		
42,5 (r = 44,0)	2,4–25,8 2500	2,4–14,3 5000	4750	3970	3400	2950	2600	2310	2070	1870	1700	1550	1420	**1300**			
40,0 (r = 41,5)	2,4–26,3 2500	2,4–14,6 5000	4840	4060	3470	3020	2660	2360	2120	1910	1740	1580	1450				
37,5 (r = 39,0)	2,4–27,1 2500	2,4–15,0 5000	5000	4200	3600	3130	2760	2450	2200	1990	1810	1650					
35,0 (r = 36,5)	2,4–27,6 2500	2,4–15,3 5000	5000	4290	3670	3200	2820	2510	2250	2040	1850						
32,5 (r = 34,0)	2,4–28,3 2500	2,4–15,7 5000	5000	4410	3780	3290	2900	2590	2320	2100							
30,0 (r = 31,5)	2,4–28,5 2500	2,4–15,8 5000	5000	4460	3820	3330	2940	2620	2350								
27,5 (r = 29,0)	2,4–27,5 2500	2,4–16,0 5000	5000	4510	3870	3370	2970	2650									
25,0 (r = 26,5)	2,4–25,0 2500	2,4–16,1 5000	5000	4550	3900	3400	3000										
22,5 (r = 24,0)	2,4–22,5 2500	2,4–16,3 5000	5000	4620	3960	3450											
20,0 (r = 21,5)	2,4–20,0 2500	2,4–16,5 5000	5000	4670	4000												

NOTE: Use working hook-radius columns across the chart. Do **not** use *jib-end r* dimension.

LIEBHERR 71 EC-B 5

QUESTION 5 — MAXIMUM LOAD (4-PART)

QUESTION 5

Parts of line: **4-part**

Jib length: **42.5 m**

Hook radius: **37.5 m**

Rigging weight: **150 kg**

Find maximum load excluding rigging.

CHART CAPACITY (4-PART)

42.5 m jib → 37.5 m radius

= 1,550 kg

(gross rated capacity)

MAXIMUM LOAD

1,550 kg − 150 kg rigging

= 1,400 kg

(load only)

ANSWER:

1,400 kg

maximum allowable load

m r	m/kg		m/kg 15,0	17,5	20,0	22,5	25,0	27,5	30,0	32,5	35,0	37,5	40,0	42,5	45,0	47,5	50,0
50,0 (r = 51,5)	2,4–22,9 **2500**	2,4–12,8 **5000**	4150	3470	2950	2560	2250	1990	1780	1600	1450	1310	1200	1090	1000	920	**850**
47,5 (r = 49,0)	2,4–24,1 **2500**	2,4–13,4 **5000**	4400	3680	3140	2730	2390	2120	1900	1710	1550	1410	1290	1180	1090	**1000**	
45,0 (r = 46,5)	2,4–25,1 **2500**	2,4–14,0 **5000**	4600	3850	3290	2860	2510	2230	2000	1800	1630	1490	1360	1250	**1150**		
42,5 (r = 44,0)	2,4–25,8 **2500**	2,4–14,3 **5000**	4750	3970	3400	2950	2600	2310	2070	1870	1700	1550	1420	**1300**			
40,0 (r = 41,5)	2,4–26,3 **2500**	2,4–14,6 **5000**	4840	4060	3470	3020	2660	2360	2120	1910	1740	1580	**1450**				
37,5 (r = 39,0)	2,4–27,1 **2500**	2,4–15,0 **5000**	5000	4200	3600	3130	2760	2450	2200	1990	1810	**1650**					
35,0 (r = 36,5)	2,4–27,6 **2500**	2,4–15,3 **5000**	5000	4290	3670	3200	2820	2510	2250	2040	**1850**						
32,5 (r = 34,0)	2,4–28,3 **2500**	2,4–15,7 **5000**	5000	4410	3780	3290	2900	2590	2320	**2100**							
30,0 (r = 31,5)	2,4–28,5 **2500**	2,4–15,8 **5000**	5000	4460	3820	3330	2940	2620	**2350**								
27,5 (r = 29,0)	2,4–27,5 **2500**	2,4–16,0 **5000**	5000	4510	3870	3370	2970	**2650**									
25,0 (r = 26,5)	2,4–25,0 **2500**	2,4–16,1 **5000**	5000	4550	3900	3400	**3000**										
22,5 (r = 24,0)	2,4–22,5 **2500**	2,4–16,3 **5000**	5000	4620	3960	**3450**											
20,0 (r = 21,5)	2,4–20,0 **2500**	2,4–16,5 **5000**	5000	4670	**4000**												

71 EC-B 5 3

42.5 m jib × 37.5 m radius = 1,550 kg rated capacity.

1,550 kg − 150 kg rigging = 1,400 kg maximum load.

LIEBHERR 71 EC-B 5

QUESTION 6 — BETWEEN LISTED RADII (2-PART)

QUESTION 6

Parts of line:	**2-part**
Jib length:	**50.0 m**
Actual radius:	**36.0 m**
Load weight:	**1,300 kg**
Rigging weight:	**100 kg**

Determine whether the lift is permitted.

RADIUS TO USE

36.0 m is not listed.

Use next larger radius: **37.5 m**

CAPACITY CHECK

Chart capacity	**1,450 kg**
Gross load	**1,400 kg**
Remaining	**50 kg**

ANSWER:

YES — PERMITTED

50 kg capacity remaining

Ausladung und Tragfähigkeit Radius and capacity / Portée et charge / Sbraccio e portata / Alcances y cargas / Alcance e capacidade de carga

m r	m/kg	15,0	17,5	20,0	22,5	25,0	27,5	30,0	32,5	35,0	37,5	40,0	42,5	45,0	47,5	50,0
50,0 (r = 51,5)	2,4–23,7 **2500**	2500	2500	2500	2500	2350	2110	1900	1730	1580	1450	1340	1240	1150	1070	**1000**
47,5 (r = 49,0)	2,4–25,0 **2500**	2500	2500	2500	2500	2500	2240	2030	1840	1690	1550	1430	1330	1230	**1150**	
45,0 (r = 46,5)	2,4–26,1 **2500**	2500	2500	2500	2500	2500	2350	2130	1940	1770	1630	1510	1400	**1300**		
42,5 (r = 44,0)	2,4–26,9 **2500**	2500	2500	2500	2500	2500	2430	2200	2010	1840	1690	1560	**1450**			
40,0 (r = 41,5)	2,4–27,4 **2500**	2500	2500	2500	2500	2500	2490	2250	2050	1880	1730	**1600**				
37,5 (r = 39,0)	2,4–28,3 **2500**	2500	2500	2500	2500	2500	2500	2340	2130	1950	**1800**					
35,0 (r = 36,5)	2,4–28,9 **2500**	2500	2500	2500	2500	2500	2500	2390	2180	**2000**						
32,5 (r = 34,0)	2,4–29,7 **2500**	2500	2500	2500	2500	2500	2500	2470	**2250**							
30,0 (r = 31,5)	2,4–30,0 **2500**	2500	2500	2500	2500	2500	2500	**2500**								
27,5 (r = 29,0)	2,4–27,5 **2500**	2500	2500	2500	2500	2500	**2500**									
25,0 (r = 26,5)	2,4–25,0 **2500**	2500	2500	2500	2500	**2500**										
22,5 (r = 24,0)	2,4–22,5 **2500**	2500	2500	2500	**2500**											
20,0 (r = 21,5)	2,4–20,0 **2500**	2500	2500	**2500**												

Actual radius 36.0 m → use next larger listed radius: 37.5 m.

1,450 kg capacity − 1,400 kg gross load = 50 kg remaining.

TEREX®

TEREX
CTT 91-5

LOAD CHART PRACTICE

READING CONFIGURATIONS

RADIUS

CAPACITY

PRACTICE. UNDERSTAND. OPERATE SAFELY.

ANSWER

34 m/min

The gross suspended load is 2.35 t.

The 2.0 t / 40 m/min line is exceeded.

Use the next higher load rating of 2.5 t.

ANSWER:
34 m/min

HOISTING MECHANISM

18 AFC 25

	t	m/min	kW	
	2,5	0 → 34	18	310 m
	2,0	0 → 40		
	1,5	0 → 51		
	1,25	0 → 60		
	1,0	0 → 70		
	5,0	0 → 17		
	4,0	0 → 20		
	3,0	0 → 26		
	2,5	0 → 30		
	2,0	0 → 35		

TEREX

QUESTION 2

ANSWER

400 V / 50 Hz or 460 V / 60 Hz

The power supply options for the 18 AFC 25 hoist mechanism are: **400 V / 50 Hz** or **460 V / 60 Hz**.

CORRECT CHOICE:
C

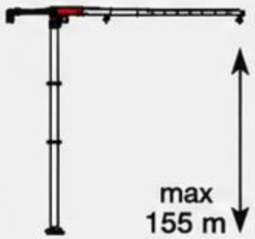
MECHANISMS

18 AFC 25

18 AFC 25	28 kVA*	400 V - 50 Hz / 460 V - 60 Hz	2000/14/CE modificata

* Average consumption · Consumo medio · Durchschnittsverbrauch · Consommation moyenne · Consumo medio · Consumo médio · Среднее потребление

HOISTING

			t	m/min	kW	
	18 AFC 25		2,5	0 → 34	18	
			2	0 → 40		
			1,5	0 → 51		
			1,25	0 → 60		
			1	0 → 70		
			5	0 → 17		
			4	0 → 20		
			3	0 → 26		
			2,5	0 → 30		
			2	0 → 35		

QUESTION 3

Jib	**50 m**
Radius	**35 m**
Row	**5 t**
Load	**1.85 t**
Rigging	**0.15 t**

CAPACITY CHECK

Gross load

1.85 + 0.15 = 2.00 t

Chart capacity

2.04 t **→ 40 kg left**

ANSWER

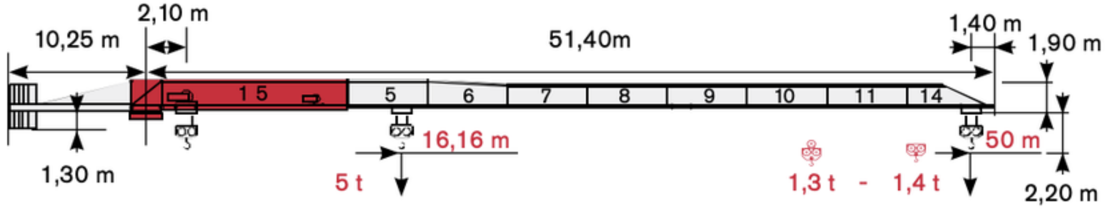
YES

40 kg remaining

LOAD DIAGRAM

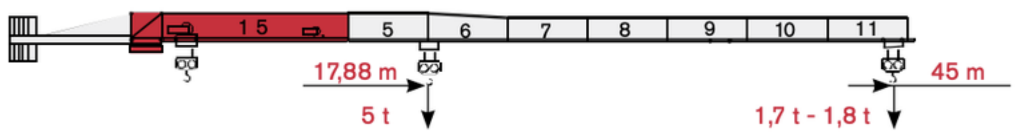
Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
2,5 t →	30,63 m	t	2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
2,5 t →	29,53 m	t	2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
5 t →	16,16 m	t	5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

50 m jib • 5 t row • 35 m radius = 2.04 t capacity

TEREX CTT 91-5 QUESTION 4 — MAXIMUM WORKING RADIUS

QUESTION 4

Jib:	**50 m**
Chart row:	**5 t**
Gross load:	**2.40 t**

Find maximum LISTED working radius.

READ OUTWARD

30 m = **2.45 t**

2.45 t ≥ 2.40 t

35 m = **2.04 t**

2.04 t < 2.40 t

So 35 m is too far.

ANSWER

30 m

maximum listed radius

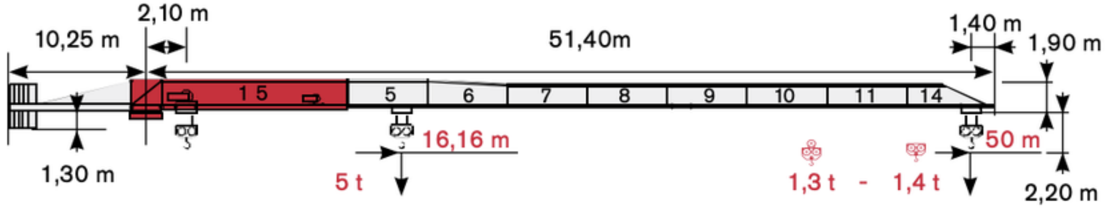
Not 16.16 m.

That is not the working radius answer.

LOAD DIAGRAM

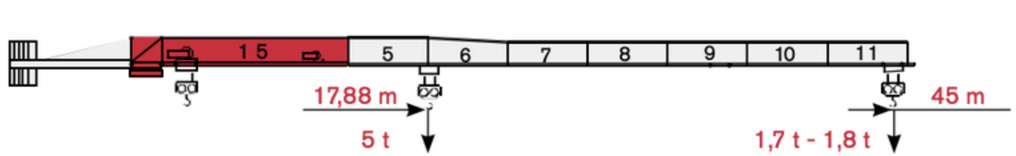
Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



		t	5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
2,5 t →	30,63 m	t	2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
2,5 t →	29,53 m	t	2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
5 t →	16,16 m	t	5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



		t	5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

5 t row: 30 m = 2.45 t; 35 m = 2.04 t. A 2.40 t gross load can reach 30 m, but not 35 m.

QUESTION 5

Jib:	50 m
Radius:	40 m
Row:	2.5 t
Crossover:	30.63 m

What is the rated capacity?

FOLLOW THE STATED ROW

2.5 t row

30.63 m crossover

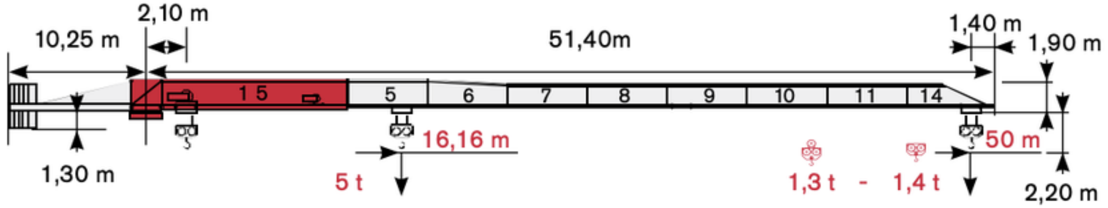
40 m radius

→ 1.83 t

LOAD DIAGRAM

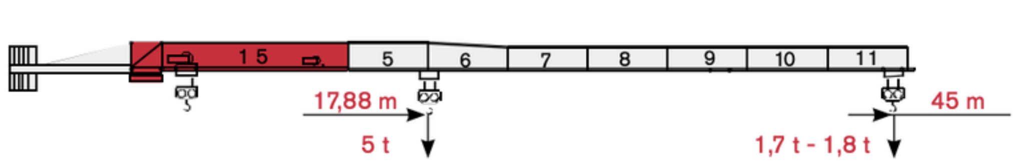
Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



				5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
	2,5 t →	30,63 m	t	2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
	2,5 t →	29,53 m	t	2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
	5 t →	16,16 m	t	5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



				5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
	2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
	2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
	5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

ANSWER

1.83 t

rated capacity

Do not use the other 2.5 t row.

50 m jib • 2.5 t / 30.63 m row • 40 m radius = 1.83 t rated capacity.

TEREX CTT 91-5

QUESTION 6 — SAME CRANE, DIFFERENT CONFIGURATION

QUESTION 6

Jib:	50 m
Radius:	40 m
Compare:	2.5 t row
	vs 5 t row

Find the capacity difference.

COMPARE AT 40 m

2.5 t / 30.63 m row	**1.83 t**
5 t / 16.16 m row	**1.73 t**

1.83 − 1.73 = 0.10 t

= 100 kg

ANSWER

0.10 t

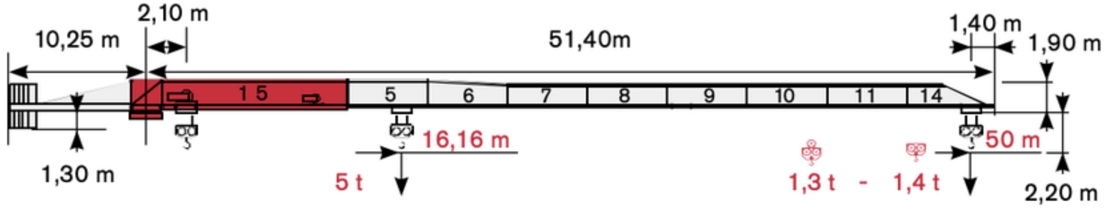
100 kg difference

Choice B

LOAD DIAGRAM

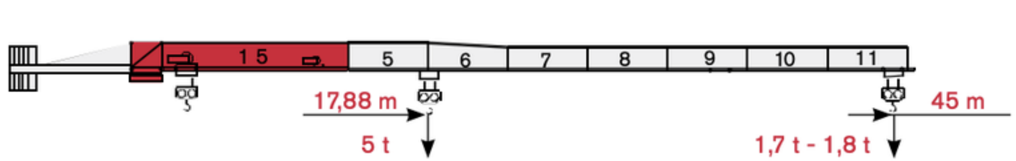
Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
2,5 t →	30,63 m	t	2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
2,5 t →	29,53 m	t	2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
5 t →	16,16 m	t	5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

At the same 40 m radius: 2.5 t row = 1.83 t; 5 t row = 1.73 t; difference = 0.10 t.

TEREX CTT 91-5 — QUESTION 7 — SHORTER JIB COMPARISON

QUESTION 7

Chart rows:	5 t
Radius:	35 m
Compare:	50 m jib
	40 m jib

CAPACITY DIFFERENCE

50 m jib	2.04 t
40 m jib	2.58 t

2.58 − 2.04 = 0.54 t

ANSWER

0.54 t

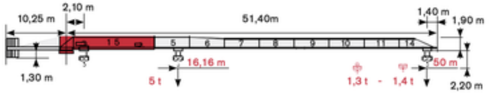
540 kg difference

50 m JIB — 5 t ROW — 35 m = 2.04 t

LOAD DIAGRAM

Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



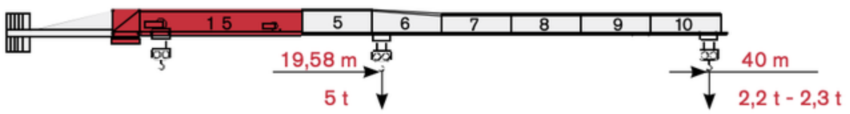
			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
2,5 t →	30,63 m	t	2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
2,5 t →	29,53 m	t	2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
5 t →	16,16 m	t	5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3

			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

40 m JIB — 5 t ROW — 35 m = 2.58 t

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m
2,5 t →	37,19 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,50	2,30
2,5 t →	35,92 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,50	2,20
5 t →	19,58 m	t	5,00	5,00	5,00	4,88	3,80	3,08	2,58	2,20

Same 35 m radius: shortening the jib from 50 m to 40 m increases capacity by 0.54 t (540 kg).

TEREX CTT 91-5 QUESTION 8 — MAXIMUM ALLOWABLE LOAD

QUESTION 8

Jib:	**45 m**
Radius:	**35 m**
Chart row:	**5 t**
Rigging:	**0.21 t**

Find load weight excluding rigging.

SUBTRACT RIGGING

Chart capacity	**2.31 t**
Rigging	**0.21 t**

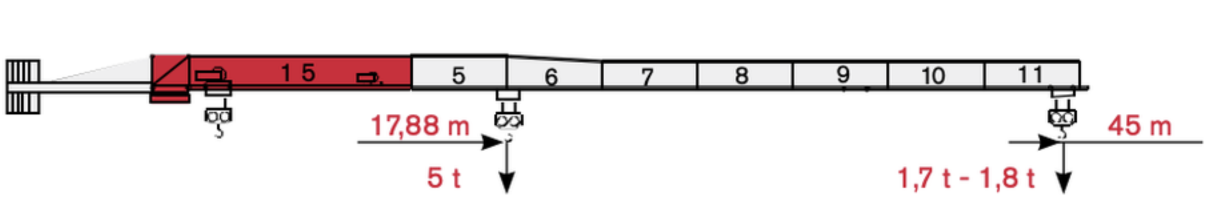
2.31 – 0.21 = 2.10 t

ANSWER

2.10 t

maximum load weight

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



		t	5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

45 m jib • 5 t row • 35 m radius = 2.31 t. Minus 0.21 t rigging = 2.10 t allowable load.

QUESTION 9

Jib:	**50 m**
Chart row:	**5 t**
Actual radius:	**32 m**
Load:	**2.10 t**
Rigging:	**0.10 t**

Is the lift permitted?

USE NEXT LISTED RADIUS

32 m is not listed.

Use 35 m.

Gross load: **2.20 t**

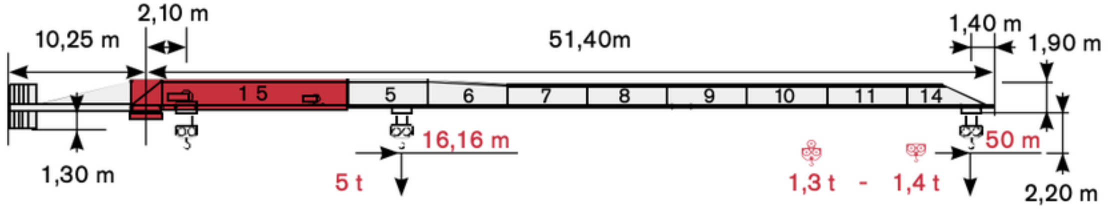
35 m capacity: **2.04 t**

Over by 0.16 t = 160 kg

LOAD DIAGRAM

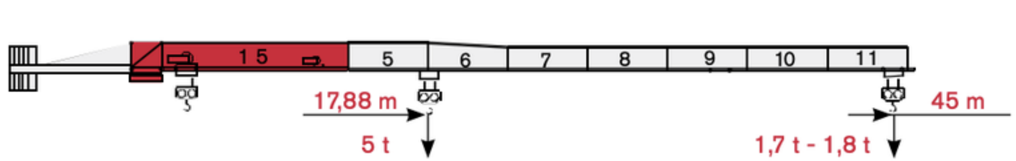
Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
2,5 t →	30,63 m	t	2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
2,5 t →	29,53 m	t	2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
5 t →	16,16 m	t	5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m
2,5 t →	33,93 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,41	2,07	1,80
2,5 t →	32,74 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,31	1,97	1,70
5 t →	17,88 m	t	5,00	5,00	5,00	4,41	3,42	2,77	2,31	1,97	1,70

ANSWER

NO

use 35 m capacity

Choice C

32 m falls between 30 m and 35 m. Use the next larger listed radius: 35 m.

2.20 t gross load > 2.04 t chart capacity, so the lift is not permitted.

TEREX CTT 91-5 QUESTION 10 — LOAD CHART + HOIST MECHANISM

QUESTION 10

Jib:	**40 m**
Radius:	**30 m**
Chart row:	**5 t**
Load:	**2.75 t**
Rigging:	**0.20 t**

STEP 1 — STRUCTURAL

Gross load: 2.75 + 0.20

= 2.95 t

Chart capacity: **3.08 t**

Remaining: **0.13 t = 130 kg**

Structurally permitted.

STEP 2 — HOIST SPEED

Gross load 2.95 t

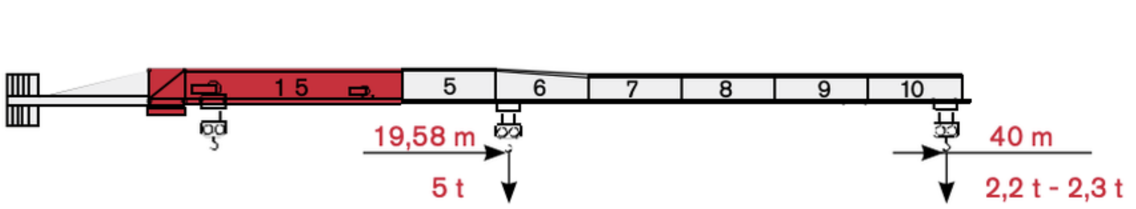
Use 3 t hoist row

MAX SPEED **26 m/min**

Answer: YES + 26 m/min

LOAD CHART: 40 m JIB • 5 t ROW • 30 m RADIUS = 3.08 t

	A (3,5 t)	C (2,3 t)
18 AFC 25	2	3



			5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m		
2,5 t →	37,19 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,50	2,30		
2,5 t →	35,92 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,50	2,20		
5 t →	19,58 m	t	5,00	5,00	5,00	4,88	3,80	3,08	2,58	2,20		

HOIST MECHANISM: 2.95 t GROSS LOAD → 3 t ROW → 26 m/min

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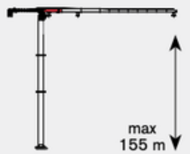
		t	m/min	kW	
18 AFC 25		2,5	0 → 34	18	310 m
		2	0 → 40		
		1,5	0 → 51		
		1,25	0 → 60		
		1	0 → 70		
		5	0 → 17		
		4	0 → 20		
		3	0 → 26		



Answer: Lift is permitted. Remaining capacity = 130 kg. Highest hoisting speed = 26 m/min.

TEREX CTT 91-5 QUESTION 11 — RELEVANT VS IRRELEVANT INFO

QUESTION 11

Jib: **35 m**

Radius: **30 m**

Chart row: **5 t**

Rigging: **0.18 t**

Ignore: 400 V / 50 Hz
28 kVA
DVF 3 4 D6

USE ONLY CAPACITY DATA

Chart capacity **3.29 t**

Rigging **0.18 t**

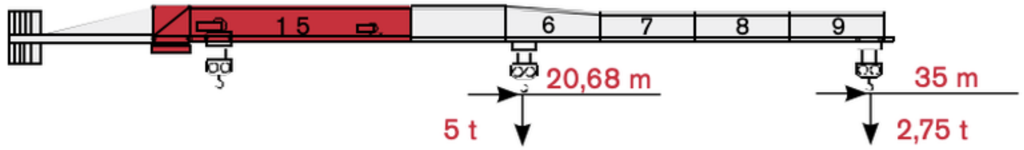
3.29 − 0.18 = 3.11 t

ANSWER

3.11 t

maximum load weight

	A (3,5 t)	C (2,3 t)
18 AFC 25	3	1



			5 m	10 m	15 m	20 m	25 m	30 m	35 m
2,5 t →	35,00 m	t	2,50	2,50	2,50	2,50	2,50	2,50	2,50
5 t →	20,68 m	t	5,00	5,00	5,00	5,00	4,04	3,29	2,75

35 m jib • 5 t row • 30 m radius = 3.29 t. Minus 0.18 t rigging = 3.11 t allowable load.

TEREX CTT 91-5 QUESTION 12 — FINAL MULTI-STEP PROBLEM

QUESTION 12

Jib:	50 m
Chart row:	5 t
Load:	2.25 t
Rigging:	0.20 t

Find radius, remaining capacity, and maximum hoisting speed.

STEP 1 — LOAD CHART

Gross load: 2.25 + 0.20 **= 2.45 t**

30 m capacity: **2.45 t**

35 m capacity: **2.04 t**

Maximum listed radius = 30 m

FINAL ANSWER

Max radius: **30 m**

Remaining: **0 kg**

Hoist speed: **30 m/min**

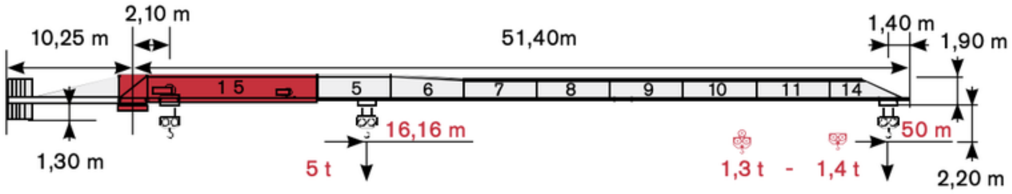
Use 2.5 t / 30 m/min row.

LOAD CHART: 50 m JIB • 5 t ROW • 30 m RADIUS = 2.45 t

LOAD DIAGRAM

Diagramma di portata · Lastkurven · Courbes de charges · Curvas de cargas · Diagrama de carga · Диаграмма грузоподъёмности

	A (3,5 t)	C (2,3 t)
18 AFC 25	4	0



				5 m	10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	50 m
2,5 t →	30,63 m	t		2,50	2,50	2 ,50	2,50	2,50	2,50	2,14	1,83	1,59	1,40
2,5 t →	29,53 m	t		2,50	2,50	2,50	2,50	2,50	2,45	2,04	1,73	1,49	1,30
5 t →	16,16 m	t		5,00	5,00	5,00	3,93	3,04	2,45	2,04	1,73	1,49	1,30

HOIST MECHANISM: 2.45 t GROSS LOAD → 2.5 t ROW → 30 m/min

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		t	m/min	kW	
18 AFC 25		2,5	0 → 34	18	310 m
		2	0 → 40		
		1,5	0 → 51		
		1,25	0 → 60		
		1	0 → 70		
		5	0 → 17		
		4	0 → 20		
		3	0 → 26		

Answer: maximum listed radius = 30 m; remaining capacity = 0 kg; maximum hoisting speed = 30 m/min.

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MDT 218 A J10

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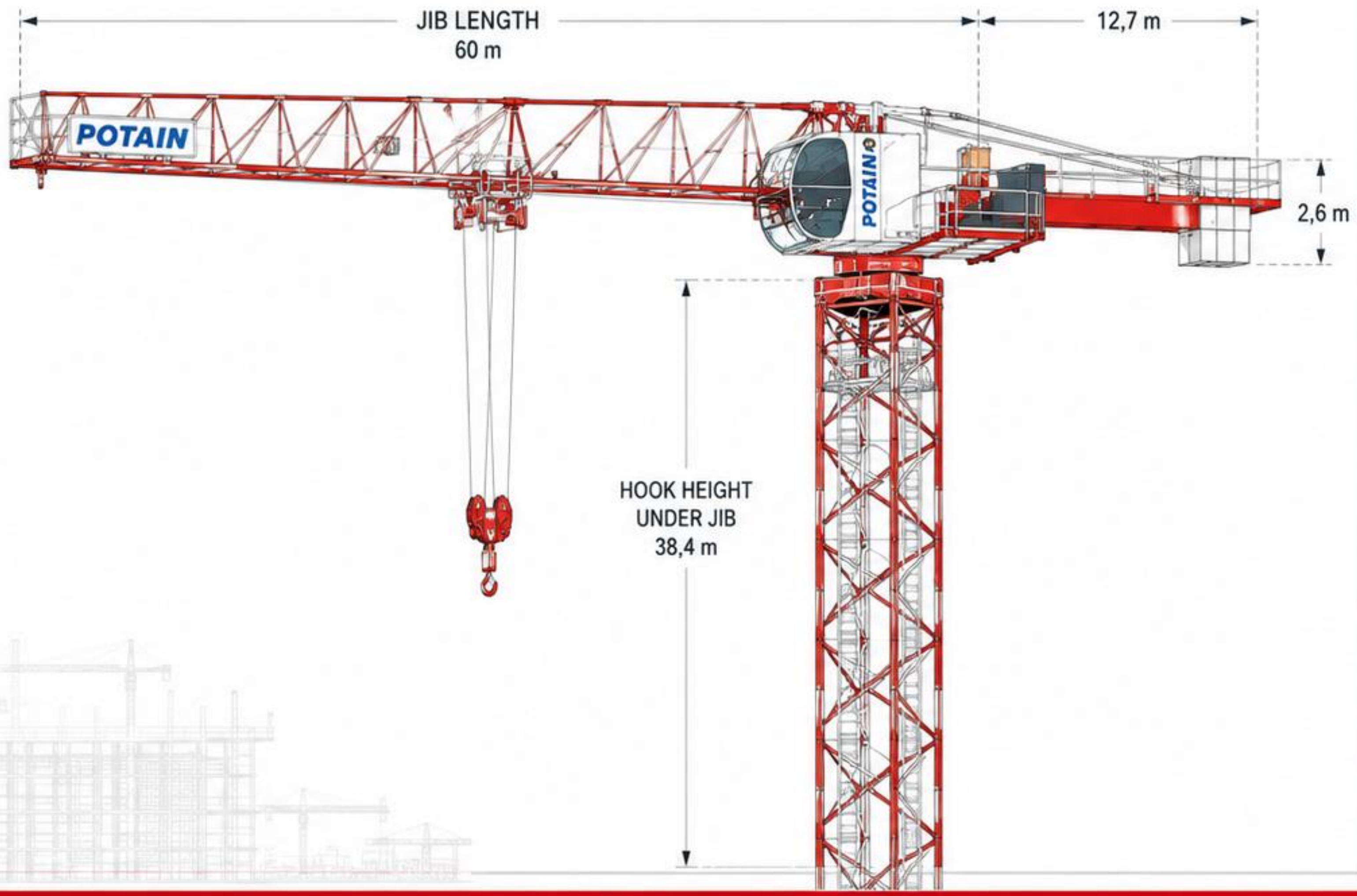

PLAN IT.
CHECK IT.
LIFT IT.

MAXIMUM WORKING RADIUS

65 m jib • Upper load chart

Load	**3.15 t**
+ Rigging	**0.20 t**
Gross suspended load	**3.35 t**

40 m radius capacity	**3.40 t**
42 m radius capacity	**3.20 t**

3.35 t fits at 40 m,
but exceeds 3.20 t at 42 m.

MAXIMUM RADIUS: 40 m

ACTUAL POTAIN LOAD CHART — 65 m JIB

Follow the radius column to the capacity row.

65 m	3	►	15	17	20	22	25	26,8	28,8	30	32	35	37	40	42	45	47	50	52	55	57	60	62	65	m
			10	8,6	7,1	6,3	5,4	5	5	4,8	4,4	4	3,7	3,4	3,2	2,95	2,8	2,6	2,45	2,3	2,2	2,05	1,95	1,85	t

READ THE CHART

40 m → 3.40 t capacity

42 m → 3.20 t capacity

Gross load = 3.35 t, so 42 m is not permitted.

The furthest listed radius that can support 3.35 t is

40 m

JIB CONFIGURATION COMPARISON

Upper load chart

Radius	**40 m**
65 m jib capacity	**3.40 t**
50 m jib capacity	**4.70 t**

Additional capacity

4.70 − 3.40 =

1.30 t

ANSWER: 1.30 t

ACTUAL POTAIN LOAD CHART — UPPER CHART

Compare the same radius across different jib configurations.

65 m	3	►	15	17	20	22	25	26,8	28,8	30	32	35	37	40	42	45	47	50	52	55
			10	8,6	7,1	6,3	5,4	5	5	4,8	4,4	4	3,7	3,4	3,2	2,95	2,8	2,6	2,45	2,
60 m	3	►		17,4	20	22	25	27	30	31,1	33,5	35	37	40	42	45	47	50	52	55
				10	8,5	7,6	6,5	5,9	5,2	5	5	4,8	4,5	4,1	3,8	3,5	3,4	3,1	3	2,
55 m	3	►		18,8	20	22	25	27	30	32	34,1	36,4	37	40	42	45	47	50	52	55
				10	9,3	8,4	7,2	6,6	5,8	5,4	5	5	4,9	4,5	4,2	3,9	3,7	3,5	3,3	3,
50 m	3	►		19,6	20	22	25	27	30	32	35	35,6	38	40	42	45	47	50	m	
				10	9,8	8,8	7,6	6,9	6,1	5,7	5,1	5	5	4,7	4,5	4,1	3,9	3,65	t	

READ ACROSS THE SAME RADIUS

65 m jib @ 40 m → 3.40 t

50 m jib @ 40 m → 4.70 t

Difference = 1.30 t additional gross capacity

MAXIMUM ALLOWABLE LOAD

60 m jib • Upper load chart

Radius	**45 m**
Rigging	**0.25 t**
Chart capacity	**3.50 t**
Minus rigging	**0.25 t**

3.50 − 0.25 =

3.25 t

ANSWER: 3.25 t

ACTUAL POTAIN LOAD CHART — UPPER CHART

Find 60 m jib, then read across to 45 m radius.

65 m	3	►	15	17	20	22	25	26,8	28,8	30	32	35	37	40	42	45	47	50	52	55	57	60	62	6
			10	8,6	7,1	6,3	5,4	5	5	4,8	4,4	4	3,7	3,4	3,2	2,95	2,8	2,6	2,45	2,3	2,2	2,05	1,95	1,
60 m	3	►		17,4	20	22	25	27	30	31,1	33,5	35	37	40	42	45	47	50	52	55	57	60	m	
				10	8,5	7,6	6,5	5,9	5,2	5	5	4,8	4,5	4,1	3,8	3,5	3,4	3,1	3	2,8	2,65	2,5	t	
55 m	3	►		18,8	20	22	25	27	30	32	34,1	36,4	37	40	42	45	47	50	52	55	m			

READ THE CAPACITY AS GROSS

60 m jib @ 45 m radius → 3.50 t

Maximum load = chart capacity − rigging

3.50 t − 0.25 t = 3.25 t

The chart gives gross rated capacity. Rigging must be deducted.

BETWEEN LISTED RADII

55 m jib • Upper load chart

Actual radius	**43.0 m**
Use listed radius	**45 m**
Load	**3.35 t**
+ Rigging	**0.20 t**
Gross load	**3.55 t**
45 m capacity	**3.90 t**
Remaining	**0.35 t**

YES — 0.35 t LEFT

ACTUAL POTAIN LOAD CHART — UPPER CHART

43.0 m is not listed, so use the next larger listed radius: 45 m.

55 m	3	▶	18,8	20	22	25	27	30	32	34,1	36,4	37	40	42	45	47	50	52	55	m
			10	9,3	8,4	7,2	6,6	5,8	5,4	5	5	4,9	4,5	4,2	3,9	3,7	3,5	3,3	3,1	t

USE THE NEXT LARGER RADIUS

43.0 m actual radius → use 45 m listed radius

55 m jib @ 45 m radius → 3.90 t capacity

3.90 t − 3.55 t gross load = 0.35 t remaining

Do not interpolate. Move to the next larger listed radius.

CORRECT CHART SELECTION

Lower supplied load chart

Jib	**50 m**
Radius	**40 m**

Read the 50 m jib row across to 40 m radius.

Gross capacity **4.50 t**

ANSWER: 4.50 t

ACTUAL POTAIN LOAD CHART — LOWER CHART

Use the lower chart stated in the question.

50 m	2,3	►	19,7	20	22	25	27	30	32	35	35,9	36,7	37	40	42	45	47	50	m
			10	9,8	8,8	7,6	7	6,2	5,7	5,2	5	5	4,9	4,5	4,3	3,9	3,7	3,45	t
45 m	2,3	►			20,2	22	25	27	30	32	35	36,9	37,7	40	42	45	m		
					[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	t		

READ THE STATED CHART

50 m jib @ 40 m radius

= 4.50 t gross capacity

This question tests chart selection: use the lower chart, not the upper chart.

CONFIGURATION DIFFERENCE

Jib	**45 m**
Radius	**40 m**
Upper chart capacity	**4.90 t**
Lower chart capacity	**4.70 t**

Difference

4.90 − 4.70 =

0.20 t

ANSWER: 0.20 t

UPPER LOAD CHART

45 m jib at 40 m radius = 4.90 t

45 m	3	▶	20,1	22	25	27	30	32	35	36,5	39	40	42	45	m
			10	9	7,8	7,1	6,3	5,9	5,3	5	5	4,9	4,6	4,25	t

LOWER LOAD CHART

45 m jib at 40 m radius = 4.70 t

45 m	2,3	▶	20,2	22	25	27	30	32	35	36,9	37,7	40	42	45	m
			10	9,1	7,9	7,2	6,4	5,9	5,3	5	5	4,7	4,4	4,05	t

Same jib and radius, different supplied chart/configuration.

Capacity difference = 0.20 t

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QUESTION 7 — ANSWER

STRUCTURAL CAPACITY CHECK

Lower load chart

Jib	**60 m**
Radius	**50 m**
Load	**2.60 t**
+ Rigging	**0.25 t**
Gross load	**2.85 t**
Chart capacity	**2.95 t**
Remaining	**0.10 t**

YES — 0.10 t LEFT

ACTUAL POTAIN LOAD CHART — LOWER CHART

Check the 60 m jib row at 50 m radius.

60 m	2,3	►	17,6	20	22	25	27	30	31,8	32,4	35	37	40	42	45	47	50	52	55	57	6
			10	8,6	7,7	6,7	6,1	5,4	5	5	4,6	4,3	3,9	3,7	3,4	3,2	2,95	2,8	2,6	2,45	2,
55 m	2,3	►	18,9	20	22	25	27	30	32	34,4	35,1	37	40	42	45	47	50	52	55	m	

COMPARE GROSS LOAD TO CHART CAPACITY

60 m jib @ 50 m radius → 2.95 t capacity

Gross load = 2.60 t + 0.25 t = 2.85 t

2.95 t − 2.85 t = 0.10 t remaining

Answer choice: A — Yes, with 0.10 t remaining.

HOIST PERFORMANCE

50 LVF 25 Optima

Load	2.25 t
+ Rigging	0.20 t
Gross suspended load	**2.45 t**

2.45 t is under the 2.5 t hoist rating.

Highest listed speed **70** m/min

ANSWER: 70 m/min

ACTUAL POTAIN MECHANISM CHART

Use the 50 LVF 25 Optima hoist row and the gross suspended load.

400 V - 50 Hz													ch - PS hp	kW	
	50 LVF 25 Optima	m/min	38	49	70	100	19	24,5	35	50	50	37	278 m
		t	5	3,75	2,5	1,25	10	7,5	5	2,5			
	50 LVF 25 GH Optima	m/min	38	49	70	100	19	24,5	35	50	50	37	515 m
		t	5	3,75	2,5	1,25	10	7,5	5	2,5			

CALCULATE GROSS LOAD FIRST

2.25 t + 0.20 t = 2.45 t gross

2.45 t fits the 2.5 t line

Highest listed speed = 70 m/min

Do not use the load-only weight. The rigging is part of the suspended load.

HOIST CONFIGURATION

Alternate / 10 t configuration

Gross suspended load **4.80 t**

50 m/min **2.5 t**

35 m/min **5.0 t**

4.80 t is too heavy for the 2.5 t / 50 line.

Use 5.0 t / 35 m/min.

ANSWER: 35 m/min

ACTUAL POTAIN HOIST MECHANISM CHART

Question 9 specifies the alternate / 10 t hoist configuration.

400 V - 50 Hz											ch - PS hp	kW	
	50 LVF 25 Optima	m/min	38	49	70	100	19	24,5	35	50	50	37	278 m
		t	5	3,75	2,5	1,25	10	7,5	5	2,5			
	50 LVF 25 GH Optima	m/min	38	49	70	100	19	24,5	35	50	50	37	515 m
		t	5	3,75	2,5	1,25	10	7,5	5	2,5			

SELECT THE HIGHEST SPEED THAT CAN CARRY 4.80 t

50 m/min → rated 2.5 t

4.80 t > 2.5 t — NOT PERMITTED

35 m/min → rated 5.0 t — PERMITTED

The hoist configuration matters: do not read the other performance group.

POWER & MECHANISM DATA

Read the manufacturer data.

Voltage	**400 V**
Frequency	**50 Hz**
Hoist demand	**58 kVA**

ANSWER B

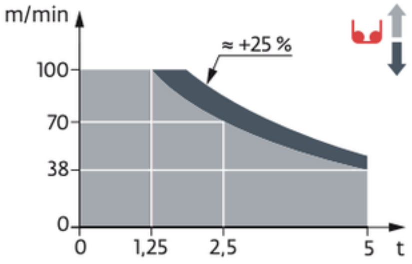
400 V • 50 Hz • 58 kVA

ACTUAL POTAIN ELECTRICAL / HOIST DATA

Use the printed supply and apparent-power values.

IEC 60204-32	kVA
400 V (+10% -10%) 50 Hz	50 LVF : 58 kVA 50 LVF GH : 58 kVA

50 LVF 25 Optima
50 LVF 25 GH Optima



400 V • 50 Hz • 58 kVA

kVA is the hoist's apparent-power requirement.

RELEVANT VS. IRRELEVANT DATA

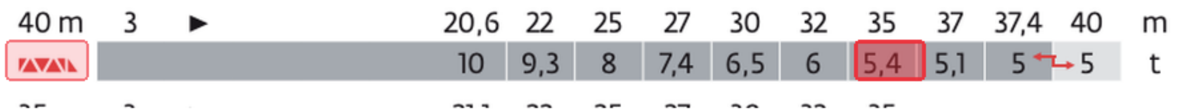
Upper load chart

Jib	**40 m**
Radius	**35 m**
Rigging	**0.20 t**
Chart capacity	**5.10 t**
- Rigging	**0.20 t**
Maximum load	**4.90 t**

ANSWER B: 4.90 t

ACTUAL POTAIN LOAD CHART — UPPER CHART

Only the jib, radius, chart configuration and rigging affect this lookup.



WHAT MATTERS FOR THIS QUESTION

40 m jib • 35 m radius • upper chart • 0.20 t rigging

IRRELEVANT TO THIS CAPACITY LOOKUP

400 V • 50 Hz • 37 kW • 7 DVF 4 trolley mechanism

5.10 t − 0.20 t = 4.90 t maximum load

Exam skill: identify the information needed for the chart lookup and ignore the rest.

RELEVANT VS. IRRELEVANT DATA

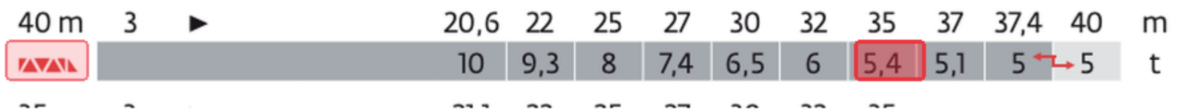
Upper load chart

Jib	**40 m**
Radius	**35 m**
Rigging	**0.20 t**
Chart capacity	**5.10 t**
- Rigging	**0.20 t**
Maximum load	**4.90 t**

ANSWER B: 4.90 t

ACTUAL POTAIN LOAD CHART — UPPER CHART

Only the jib, radius, chart configuration and rigging affect this lookup.



WHAT MATTERS FOR THIS QUESTION

40 m jib • 35 m radius • upper chart • 0.20 t rigging

IRRELEVANT TO THIS CAPACITY LOOKUP

400 V • 50 Hz • 37 kW • 7 DVF 4 trolley mechanism

5.10 t − 0.20 t = 4.90 t maximum load

Exam skill: identify the information needed for the chart lookup and ignore the rest.

FINAL MULTI-CHART PROBLEM

65 m jib • 45 m radius

Load	2.45 t
+ Rigging	0.20 t
Gross load	**2.65 t**
Structural capacity	**2.80 t**
Remaining	**0.15 t**
Hoist speed	**49** m/min

PERMITTED

0.15 t left • 49 m/min

1 STRUCTURAL LOAD CHART

65 m	3	▶	15	17	20	22	25	26,8	28,8	30	32	35	37	40	42	45	47	50	52	55	5
			10	8,6	7,1	6,3	5,4	5	5	4,8	4,4	4	3,7	3,4	3,2	2,95	2,8	2,6	2,45	2,3	2,

65 m jib @ 45 m radius = 2.80 t

2 HOIST PERFORMANCE

400 V - 50 Hz						
50 LVF 25 Optima	m/min	38	49	70	100	19
	t	5	3,75	2,5	1,25	10
50 LVF 25 GH	m/min	38	49	70	100	19

2.65 t is greater than the 2.5 t rating at 70 m/min.

Next usable rating: 3.75 t at 49 m/min.

FINAL: PERMITTED • 0.15 t REMAINING • 49 m/min